6 Elementary Differential Geometry

In this section, we would like to introduce the *coordinate-based* approach to the differential geometry, which is always called the *classical differential geometry*. Calculations are always associated with the tensor calculus. It is the main mathematical tools used in the introductory course on general relativity. The *abstract* approaches and the aspects of fibre bundles, i.e., the *modern differential geometry*, could be found in Appendix. B.

6.1 Curves on $\mathbb{R}^2$

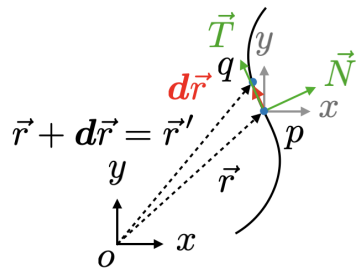


Fig. 6.1: A curve on $\mathbb{R}^2$.

Consider 2-dimensional Euclidean space, which is an $\mathbb{R}^2$ plane, which is constructed by a *fixed* frame. So any point in this plane can be described by a set coordinate function with respect to a reference point o, a 2-dimensional *Cartesian coordinate system*. A curve can be realized by a time series set of points, and hence, we parameterize a curve as a vector-valued function of one variable $t \in \mathbb{R}$, i.e., $\vec{r}(t) = \big(x(t), y(t)\big)^T$. The *velocity* vector at point $\vec{r}$ is given by

$$\vec{v}(t) = \dot{\vec{r}}(t) := \frac{\boldsymbol{d}\vec{r}}{dt} = e_x \frac{dx}{dt} + e_y \frac{dy}{dt} = \big(\dot{x}(t), \dot{y}(t)\big)^T, \tag{6.1}$$

where $\boldsymbol{d}\vec{r} = e_x\, dx + e_y\, dy + e_z\, dz$ and the *norm* of $\vec{v}$ is

$$\|\vec{v}\,\| = \|\dot{\vec{r}}(t)\| = \sqrt{\dot{\vec{r}} \cdot \dot{\vec{r}}} = \sqrt{\dot{x}^2 + \dot{y}^2}\,, \tag{6.2}$$

where $\dot{x} := dx/dt$. Then the arc length between a interval $[a, t]$ can be regarded as a function of parameter t given by

$$s = \int_a^t \|\dot{\vec{r}}(t')\|\, dt' \equiv f(t). \tag{6.3}$$

It can be shown that by using s as a parameter of a curve $\vec{r}(t(s))$ with $t = f^{-1}(s)$, the norm of the velocity is normalized, i.e.,

$$\|\vec{r}\,'(s)\| = \sqrt{x'^2 + y'^2} = \sqrt{\dot{x}^2 + \dot{y}^2}\, \frac{dt}{ds} = \|\dot{\vec{r}}\,\| \frac{dt}{ds} = \frac{df}{dt}\frac{dt}{ds} = 1, \tag{6.4}$$

where $x' := dx/ds$. Therefore, we can define a *unit tangent vector* as a *velocity* vector parameterized by s

$$\boxed{\vec{T} := \vec{r}\,'(s).} \tag{6.5}$$

By differentiate $\vec{r}\cdot\vec{r}=1$ with respect to s again, we obtain

$$\vec{r}''\cdot\vec{r}'+\vec{r}'\cdot\vec{r}''=0 \qquad\Longrightarrow\qquad \vec{r}''\cdot\vec{r}'=0, \tag{6.6}$$

so the *acceleration* vector $\vec{a}(s)\coloneqq\vec{r}''=\vec{T}'$ is normal to $\vec{T}=\vec{r}'$. Due to the orthonormality, we can define a *principal normal vector*

$$\boxed{\vec{N}=\frac{\vec{r}''}{\|\vec{r}''\|},} \tag{6.7}$$

which is a *unit* normal vector. We can rewrite (6.7) as

$$\boxed{\vec{T}'=\kappa(s)\,\vec{N}} \tag{6.8}$$

related to the normal vector $\vec{N}$, which spans a space normal to the tangent space spanned by $\vec{T}$ at any point of the curve $\vec{r}(s)$. The factor

$$\boxed{\kappa(s)\coloneqq\|\vec{r}''\|=\|\vec{a}(s)\|}>0 \tag{6.9}$$

as the *magnitude* of the acceleration vector is defined as the *curvature* of the curve $\vec{r}(s)$ at s.

Kinematically, the curvature of a space can be realized by the motion of a particle:

$$\text{Curvature}\sim\text{Acceleration}. \tag{6.10}$$

Now we can use $\vec{T}$ and $\vec{N}$ as a frame $(\vec{r};\vec{T},\vec{N})$ to span a two-dimensional space which is called the *osculating plane*.

Because of $\vec{T}\cdot\vec{N}=0$, we can differentiate the relation and obtain a new equation

$$0=\vec{T}'\cdot\vec{N}+\vec{T}\cdot\vec{N}'=\kappa\vec{N}\cdot\vec{N}+\vec{T}\cdot\vec{N}' \qquad\Longrightarrow\qquad \vec{T}\cdot\vec{N}'=-\kappa \quad\text{or}\quad \boxed{\vec{N}'=-\kappa\vec{T},} \tag{6.11}$$

where $\vec{N}'$ can be uniquely determined by $\vec{T}$ due to $\vec{N}'\cdot\vec{N}=0$. As a result, we have the derivation equations of a frame $(\vec{r};\vec{T},\vec{N})$:

$$\begin{cases}\vec{r}'=\quad+\vec{T},\\ \vec{T}'=\qquad\quad+\kappa\vec{N},\\ \vec{N}'=-\kappa\vec{T},\end{cases}\qquad\Longrightarrow\qquad \begin{pmatrix}\vec{r}'\\ \vec{T}'\\ \vec{N}'\end{pmatrix}=\begin{pmatrix}1&0\\0&\kappa\\-\kappa&0\end{pmatrix}\begin{pmatrix}\vec{T}\\ \vec{N}\end{pmatrix}, \tag{6.12}$$

which is called the *Frenet-Serret formula* in $\mathbb{R}^2$.

Gauss map. Gauss map G of a curve is a mapping which globally send all the points p of a curve to an *unit circle* S^1 (a Gauss circle) centered at C and send the corresponding normal vector $\vec{N}$ to a radius vector from C pointing to S^1, which is shown as Fig. 6.2. Therefore, $\vec{N}$ can be represented as a position vector of a point on S^1.

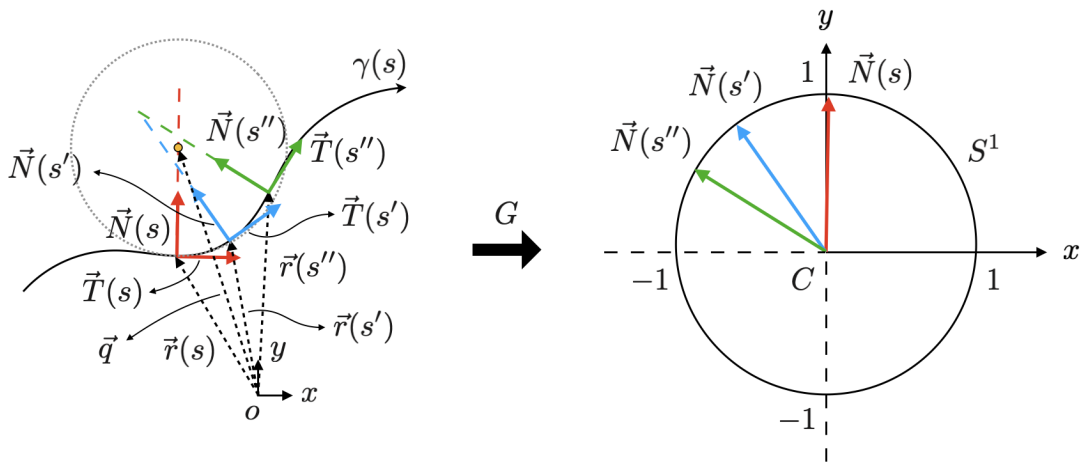


Fig. 6.2: The Gauss map G of a curve.

It is interesting to find a correspondence between the curve γ and the Gauss circle S^1:

$$\vec{r} \text{ of a curve } \gamma \ \sim\ \vec{N} \text{ of } S^1 \quad \Longrightarrow \quad \boldsymbol{d}\vec{r} \text{ of a curve } \gamma \ \sim\ \boldsymbol{d}\vec{N} \text{ of } S^1. \tag{6.13}$$

Let's consider two normal vectors $\vec{N}(s)$ and $\vec{N}(s')$ with respect to two infinitesimally nearby points $\vec{r}(s)$ and $\vec{r}(s')$, where $s' = s + ds$ is infinitesimally close to s. We can expand $\vec{N}(s')$ at s:

$$\begin{aligned}
\vec{N}(s') &= \vec{N}(s + ds) \\
&\approx \vec{N}(s) + \vec{N}'(s)ds \\
&= \vec{N}(s) + \big(-\kappa(s)\vec{T}(s)\big)ds \\
&= \vec{N}(s) + \big(-\kappa(s)\,ds\big)\vec{T}(s)\,,
\end{aligned} \tag{6.14}$$

which is the parametrization of a point under the Gauss map. Thus, we know the distance between two infinitesimal points represented by $\vec{N}(s)$ and $\vec{N}(s')$ on Gauss circle given by

$$\|\vec{N}(s') - \vec{N}(s)\| = \|\boldsymbol{d}\vec{N}\| = \kappa(s)\,ds\,. \tag{6.15}$$

And we also have the displacement vector

$$\boldsymbol{d}\vec{r} \equiv \vec{r}(s') - \vec{r}(s) \approx \big(\cancel{\vec{r}(s)} + \vec{r}\,'(s)\,ds\big) - \cancel{\vec{r}(s)} = \vec{T}(s)\,ds \tag{6.16}$$

with the norm of

$$\|\boldsymbol{d}\vec{r}\| = \|\vec{r}(s') - \vec{r}(s)\| \approx ds\,. \tag{6.17}$$

Therefore, in the *local* region, the ratio of the length between two points on the Gauss circle and curve, i.e., $\|\boldsymbol{d}\vec{N}\|/\|\boldsymbol{d}\vec{r}\|$ can be calculate by

$$\frac{\|\vec{N}(s') - \vec{N}(s)\|}{\|\vec{r}(s') - \vec{r}(s)\|} = \frac{\|\boldsymbol{d}\vec{N}\|}{\|\boldsymbol{d}\vec{r}\|} \approx \frac{\kappa(s)\,ds}{ds} = \kappa(s)\,, \tag{6.18}$$

which measure the curvature of a curve, $\kappa(s)$, at the neighborhood of a local point $\vec{r}$.

According to the example of a circle, the curvature is $\kappa = 1/r$. So, we can assume that the position vector of the center of osculating circle is identified by the the vector

$$\vec{q} = \vec{r} + r\vec{N} = \vec{r} + (1/\kappa)\vec{N}. \tag{6.19}$$

Taking the derivative of $\vec{q}$ with respect to s, we obtain

$$\vec{q}\,' = \vec{r}\,' + \frac{1}{\kappa}\vec{N}' = \vec{T} + \frac{1}{\kappa}(-\kappa\vec{T}) = 0\,, \tag{6.20}$$

which means that $\vec{q}$ is fixed, i.e., $\vec{q}$ points to the center of the osculating circle with radius $1/\kappa$. By considering the Gauss map of a circle. The radius vector of the the Gauss circle should be $\vec{N}$, and its center C corresponds to the center of the osculating circle with position vector $\vec{q}$. Thus, in the case of a circle, the Gauss circle can be imaged as osculating circle by rescaling the length of the radius to unity.

6.2 Curves in $\mathbb{R}^3$

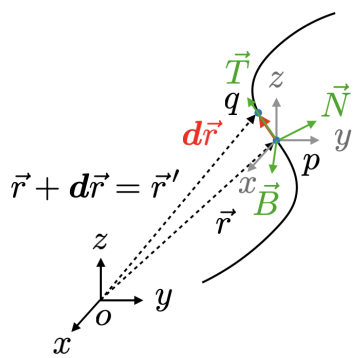


Fig. 6.3: A curve in $\mathbb{R}^3$.

In $\mathbb{R}^3$ we can similarly define the tangent vector $\vec{T}$ and the principal normal vector $\vec{N}$ of a point on a curve. In addition, we can define the so-called *binormal vector* through the *vector product* of $\vec{T}$ and $\vec{N}$

$$\boxed{\vec{B} = \vec{T} \times \vec{N}} \quad \text{or} \quad \boxed{\vec{B} = \vec{T} \wedge \vec{N}.}^{\dagger 13} \tag{6.21}$$

Again, we can compute the derivative of $\vec{N}$ to find the relationship between each component, i.e., the *Frenet-Serret formula* in $\mathbb{R}^3$. By differetiating the orthogonality conditions

$$\begin{cases} \vec{T} \cdot \vec{N} = 0, & (6.22a) \\ \vec{T} \cdot \vec{B} = 0, & (6.22b) \\ \vec{N} \cdot \vec{B} = 0, & (6.22c) \end{cases}$$

[†13] The *cross product* $\times$ of two vectors cannot be used in higher dimensional space, generally we can use the *exterior product* $\wedge$ for any dimensions $n \geq 3$.

we obtain

$$\begin{cases} 0 = \underbrace{\vec{T}\,'}_{\kappa\vec{N}} \cdot \vec{N} + \vec{T} \cdot \vec{N}\,' = \kappa + \vec{T} \cdot \vec{N}\,' & \text{(6.23a)} \\ 0 = \underbrace{\vec{T}\,'}_{\kappa\vec{N}} \cdot \vec{B} + \vec{T} \cdot \vec{B}\,' = 0 + \vec{T} \cdot \vec{B}\,', & \text{(6.23b)} \\ 0 = \vec{N}\,' \cdot \vec{B} + \vec{N} \cdot \vec{B}\,'. & \text{(6.23c)} \end{cases}$$

Because $\{\vec{T}, \vec{N}, \vec{B}\}$ is an orthonormal frame, then we can assume that

$$\vec{N}' = a\vec{T} + b\vec{B} \quad \text{and} \quad \vec{B}' = c\vec{T} + d\vec{N}. \tag{6.24}$$

By the relations (6.23) above, we have

$$a = -\kappa, \quad c = 0 \quad \text{and} \quad d = -b. \tag{6.25}$$

Consequently $\vec{N}\,'$ and $\vec{B}\,'$ are respectively written by

$$\vec{N}\,' = -\kappa\vec{T} + b\vec{B} \quad \text{and} \quad \vec{B}\,' = -\tau\vec{N}, \tag{6.26}$$

where $\tau(s) \coloneqq b(s)$ is defined as the *torsion* of a curve. Then the Frenet-Serret formula is

$$\begin{cases} \vec{r}\,' = +\vec{T}, \\ \vec{T}\,' = \qquad +\kappa\vec{N}, \\ \vec{N}\,' = -\kappa\vec{T} \qquad +\tau\vec{B}, \\ \vec{B}\,' = \qquad -\tau\vec{N}, \end{cases} \implies \begin{pmatrix} \vec{r}\,' \\ \vec{T}\,' \\ \vec{N}\,' \\ \vec{B}\,' \end{pmatrix} = \left(\begin{array}{cc:c} 1 & 0 & 0 \\ 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ \hdashline 0 & -\tau & 0 \end{array}\right) \begin{pmatrix} \vec{T} \\ \vec{N} \\ \vec{B} \end{pmatrix}. \tag{6.27}$$

6.3 Surface Theory in $\mathbb{R}^3$

In this section we consider a 2-dimensional surface $\mathcal{M}$ realized as a set of points in $\mathbb{R}^3$, i.e., $\mathcal{M}$ is a hypersurface (codimension one) of $\mathbb{R}^3$. Any point p of a surface is represented by a position vector

$$\vec{r}(u, v) = \vec{e}_x\, x(u, v) + \vec{e}_y\, y(u, v) + \vec{e}_z\, z(u, v), \tag{6.28}$$

which parameterized by *two* variables u and v, as a vector-valued function of two variables in $\mathbb{R}^3$. Similar to the previous situation, we have p and q in $\mathbb{R}^3$, however, two points are restricted on $\mathcal{M}$ as shown in Fig. 6.4. The infinitesimal displacement vector is a *two-dimensional* vector tangent to $\mathcal{M}$ in $\mathbb{R}^3$, which can always be written as

$$\boldsymbol{d}\vec{r} = \vec{e}_x\, dx + \vec{e}_y\, dy + \vec{e}_z\, dz \equiv \boldsymbol{d}\vec{r}_{(e)}. \tag{6.29}$$

We call the expression the *local extension* of $\boldsymbol{d}\vec{r}$ to $\mathbb{R}^3$. This local extension is a higher dimensional representation particularly denoted by $\boldsymbol{d}\vec{r}_{(e)}$. Such a infinitesimal displacement vector can alternatively be written by

$$\boldsymbol{d}\vec{r} = \vec{e}_x \left(\frac{\partial x}{\partial u}\, du + \frac{\partial x}{\partial v}\, dv \right) + \vec{e}_y \left(\frac{\partial x}{\partial u}\, du + \frac{\partial y}{\partial v}\, dv \right) + \vec{e}_z \left(\frac{\partial x}{\partial u}\, du + \frac{\partial z}{\partial v}\, dv \right) \tag{6.30}$$

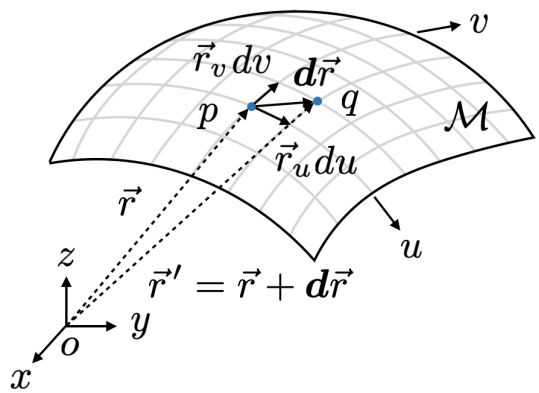


Fig. 6.4: An infinitesimal displacement vector in $\mathbb{R}^3$.

since

$$\boxed{dx^i = \frac{\partial x^i}{\partial u^a}\,du^a \quad (i, j = x, y, z \text{ and } a, b = u, v),} \tag{6.31}$$

or more precisely,

$$\begin{cases} dx = \dfrac{\partial x}{\partial u}\,du + \dfrac{\partial x}{\partial v}\,dv, & (6.32\text{a}) \\ dy = \dfrac{\partial y}{\partial u}\,du + \dfrac{\partial y}{\partial v}\,dv, & (6.32\text{b}) \\ dz = \dfrac{\partial z}{\partial u}\,du + \dfrac{\partial z}{\partial v}\,dv. & (6.32\text{c}) \end{cases}$$

The infinitesimal displacement vector through a point $p = (u, v)$ along the u-direction (with a fixed v) can be *linearly* approximated by

$$d\vec{r}_u := \vec{r}(u + du, v) - \vec{r}(u, v) \approx \cancel{\vec{r}(u, v)} + \frac{\partial \vec{r}(u, v)}{\partial u}\,du - \cancel{\vec{r}(u, v)} := \vec{r}_u\,du, \tag{6.33}$$

which is the infinitesimal vector with a small segment on the so-called *u-curves*. Here we have the definition

$$\vec{r}_u := \frac{\partial \vec{r}(u, v)}{\partial u} = \vec{e}_x\,\frac{\partial x}{\partial u} + \vec{e}_y\,\frac{\partial y}{\partial u} + \vec{e}_z\,\frac{\partial z}{\partial u}, \tag{6.34}$$

where we note that $\{e_x, e_y, e_z\}$ is a *fixed* (or an *invariant*) frame, its derivative is vanished:

$$\frac{\partial \vec{e}_x}{\partial u} = \frac{\partial \vec{e}_y}{\partial u} = \frac{\partial \vec{e}_z}{\partial u} = 0. \tag{6.35}$$

Similarly, we have the displacement vector through the point p along the v-direction (with a fixed u) on the v-curves

$$d\vec{r}_v \approx \vec{r}_v\,dv \quad \text{with} \quad \vec{r}_v := \frac{\partial \vec{r}(u, v)}{\partial v} = \vec{e}_x\,\frac{\partial x}{\partial v} + \vec{e}_y\,\frac{\partial y}{\partial v} + \vec{e}_z\,\frac{\partial z}{\partial v}. \tag{6.36}$$

Therefore, we obtain an approximation

$$\boxed{d\vec{r} = d\vec{r}_u + d\vec{r}_v = \vec{r}_u\,du + \vec{r}_v\,dv = \vec{r}_a\,du^a,} \tag{6.37}$$

which is illustrated by Fig. 6.5. It is obvious that the *linearly* approximated equation above is the same as (6.29). In addition the infinitesimal displacement vector $d\vec{r}$ is tangent at the point p, we can consequently identify the set of vector $\{\vec{r}_u, \vec{r}_v\}$ as a basis of the tangent space at p on $\mathcal{M}$.

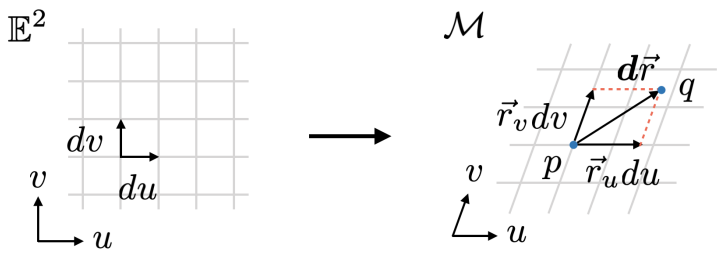


Fig. 6.5: A local coordinates (u, v) and an infinitesimal displacement vector $\boldsymbol{d}\vec{r}$ on $\mathcal{M}$.

A curve on a surface. Furthermore, if coordinates are functions of one parameter t or s, respectively, a curve on $\mathcal{M}$ can be obtained and parameterized either by $\gamma(t) = \vec{r}\big(x(t), y(t), z(t)\big)$ or $\gamma(s) = \vec{r}\big(u(s), v(s)\big)$. The corresponding tangent vector is respectively obtained by either

$$\dot{\vec{r}}_{(e)} = \vec{e}_x\,\dot{x} + \vec{e}_y\,\dot{y} + \vec{e}_z\,\dot{z} \quad \text{or} \quad \vec{r}\,' = \vec{r}_u\,u' + \vec{r}_v\,v'. \tag{6.38}$$

First fundamental (quadratic) form. We consider the norm of the infinitesimal displacement vector on $\mathcal{M}$ by using (6.37)

$$\begin{aligned} ds^2 &= \|\boldsymbol{d}\vec{r}\,\|^2 = \boldsymbol{d}\vec{r}\cdot\boldsymbol{d}\vec{r} \\ &= \underbrace{\vec{r}_u\cdot\vec{r}_u}_{E}\,du\,du + \underbrace{\vec{r}_u\cdot\vec{r}_v}_{F}\,du\,dv + \underbrace{\vec{r}_v\cdot\vec{r}_u}_{F}\,dv\,du + \underbrace{\vec{r}_v\cdot\vec{r}_v}_{G}\,dv\,dv \\ &= E\,du\,du + 2F\,du\,dv + G\,du\,dv \\ &= \begin{pmatrix} du & dv \end{pmatrix} \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} du \\ dv \end{pmatrix}. \end{aligned} \tag{6.39}$$

where

$$E = \left(\frac{\partial x}{\partial u}\right)^2 + \left(\frac{\partial y}{\partial u}\right)^2 + \left(\frac{\partial z}{\partial u}\right)^2, \tag{6.40a}$$

$$F = \frac{\partial x}{\partial u}\frac{\partial x}{\partial v} + \frac{\partial y}{\partial u}\frac{\partial y}{\partial v} + \frac{\partial z}{\partial u}\frac{\partial z}{\partial v}, \tag{6.40b}$$

$$G = \left(\frac{\partial x}{\partial v}\right)^2 + \left(\frac{\partial y}{\partial v}\right)^2 + \left(\frac{\partial z}{\partial v}\right)^2. \tag{6.40c}$$

Define dS^2 to be I, the *first fundamental* (*quadratic*) *form* of $\mathcal{M}$:

$$\boxed{\mathrm{I} := ds^2 = g_{ab}(u, v)\,du^a du^b \quad (a, b = u, v).} \tag{6.41}$$

In addition, we have local extension $d\vec{r}_{(e)}$, hence,

$$\begin{aligned} ds^2 &= (e_i\,dx^i)\cdot(e_j\,dx^j) = g_{ij}\,dx^i dx^j \\ &= g_{ij}\left(\frac{\partial x^i}{\partial u^a}du^a\right)\left(\frac{\partial x^j}{\partial u^b}du^b\right) = \left(g_{ij}\frac{\partial x^i}{\partial u^a}\frac{\partial x^j}{\partial u^b}\right)du^a du^b. \end{aligned} \tag{6.42}$$

Then we obtain the component of the metric tensor on $\mathcal{M}$

$$\boxed{g_{ab}(u, v) := g_{ij}(x, y, z)\frac{\partial x^i}{\partial u^a}\frac{\partial x^j}{\partial u^b} \quad (i, j = x, y, z \text{ and } a, b = u, v)} \tag{6.43}$$

called the *pullback of* $g_{ij}(x, y, z)$.

> **Note:** In *usual* cases, we will concern n-dimensional space $\mathcal{M}$ only and do not take $\mathcal{M}$ as a subspace of a m-dimensional Euclidean space $\mathbb{R}^m$. Therefore, there is no higher dimensional representation of the local extension of $\boldsymbol{d}\vec{r}$ to $\mathbb{R}^m$.

Orthonomal frame. By $\{\vec{r}_u, \vec{r}_v\}$, we can find a set as an orthonormal frame at p on $\mathcal{M}$ by the *Gram-Schmit* procedure. Define a operator projecting out the $\vec{e}_a$ component

$$\boxed{\vec{r}_a^{\perp} = \vec{r}_a - \sum_{b=1}^{a-1} g\big(\vec{r}_a, \vec{e}_b\big)\vec{e}_b \quad \text{and} \quad \vec{e}_b := \frac{\vec{r}_b^{\perp}}{\|\vec{r}_b^{\perp}\|}.} \tag{6.44}$$

So the first and second basis vectors are

$$\vec{r}_u \longmapsto \vec{e}_1 := \frac{\vec{r}_u^{\perp}}{\|\vec{r}_u^{\perp}\|} = \frac{\vec{r}_u}{\|\vec{r}_u\|} \tag{6.45}$$

and

$$\vec{r}_v \longmapsto \vec{e}_2 := \frac{\vec{r}_v^{\perp}}{\|\vec{r}_v^{\perp}\|} = \frac{\vec{r}_v - (\vec{r}_v \cdot \vec{e}_1)\vec{e}_1}{\|\vec{r}_v - (\vec{r}_v \cdot \vec{e}_1)\vec{e}_1\|}\,, \tag{6.46}$$

Then, we can easily obtain the *unit normal vector* of $\mathcal{M}$

$$\boxed{\vec{e}_3 = \vec{e}_1 \wedge \vec{e}_2 := \vec{n}(u, v)} \tag{6.47}$$

and its differential $d\vec{n}$ is

$$\boxed{\boldsymbol{d}\vec{n} = \frac{\partial \vec{n}}{\partial u} du + \frac{\partial \vec{n}}{\partial v} dv = \vec{n}_u\, du + \vec{n}_v\, dv = \vec{n}_a\, du^a,} \tag{6.48}$$

where we have $\vec{n}_a := \vec{n}_{,a}$. In addition the unit normal vector $\vec{n}$ can also be obtained by

$$\boxed{\vec{n}(u, v) = \frac{\vec{r}_u \wedge \vec{r}_v}{\|\vec{r}_u \wedge \vec{r}_v\|}.} \tag{6.49}$$

Therefore, we have two sets of complete basis vector of frame $(p; \vec{r}_u, \vec{r}_u, \vec{n})$ or $(p; \vec{e}_1, \vec{e}_2, \vec{e}_3)$ in $\mathbb{R}^3$. Particularly, the frame $(p; \vec{e}_1, \vec{e}_2, \vec{e}_3)$ is an *orthonormal frame.*

Second fundamental (quadratic) form. The frame $(p; \vec{r}_u, \vec{r}_u, \vec{n})$ is shown by Fig. 6.6, and we will be concentrated on such a frame for the following calculations. Vectors $\vec{r}_u$ and $\vec{r}_v$ are tangent to $\mathcal{M}$, then by relations of $\vec{n} \cdot \vec{r}_a = 0$ $(a = u, v)$, we obtain the derivative equations with respect to u^b $(b = u, v)$:

$$\underbrace{\vec{n} \cdot \vec{r}_{ab}}_{b_{ab}} + \vec{n}_b \cdot \vec{r}_a = 0 \qquad \Longrightarrow \qquad \boxed{b_{ab} := \vec{n} \cdot \vec{r}_{ab} = -\vec{r}_a \cdot \vec{n}_b.} \tag{6.50}$$

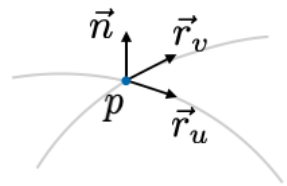


Fig. 6.6: A local frame of $\mathcal{M}$.

More precisely equations are

$$\begin{cases} \vec{n}\cdot\vec{r}_{uu} = -\vec{r}_u\cdot\vec{n}_u := L, & (6.51a)\\ \vec{n}\cdot\vec{r}_{uv} = -\vec{r}_u\cdot\vec{n}_v := M, & (6.51b)\\ \vec{n}\cdot\vec{r}_{vu} = -\vec{r}_v\cdot\vec{n}_u := M, & (6.51c)\\ \vec{n}\cdot\vec{r}_{vv} = -\vec{r}_v\cdot\vec{n}_v := N. & (6.51d)\end{cases}$$

We compute a quadratic form $-\boldsymbol{d}\vec{r}\cdot\boldsymbol{d}\vec{n}$ by using (6.37), (6.48) and (6.51), then it is read as

$$\begin{aligned} -\boldsymbol{d}\vec{r}\cdot\boldsymbol{d}\vec{n} &= -(\vec{r}_u\,du+\vec{r}_v\,dv)(\vec{n}_u\,du+\vec{n}_v\,dv)\\ &= -(\vec{r}_u\cdot\vec{n}_u\,du\,du+\vec{r}_u\cdot\vec{n}_v\,du\,dv+\vec{r}_v\cdot\vec{n}_u\,dv\,du+\vec{r}_v\cdot\vec{n}_v\,dv\,dv)\\ &= \underbrace{\vec{n}\cdot\vec{r}_{uu}}_{L}\,du\,du+\underbrace{\vec{n}\cdot\vec{r}_{uv}}_{M}\,du\,dv+\underbrace{\vec{n}\cdot\vec{r}_{vu}}_{M}\,dv\,du+\underbrace{\vec{n}\cdot\vec{r}_{vv}}_{N}\,dv\,dv\\ &= L\,du\,du+2M\,du\,dv+N\,du\,dv \qquad (6.52a)\\ &= \begin{pmatrix} du & dv\end{pmatrix}\begin{pmatrix} L & M\\ M & N\end{pmatrix}\begin{pmatrix} du\\ dv\end{pmatrix} \qquad (6.52b)\end{aligned}$$

Now the second fundamental (quadratic) form is defined by

$$\boxed{\mathrm{II} := -\boldsymbol{d}\vec{r}\cdot\boldsymbol{d}\vec{n} = \vec{n}\cdot\boldsymbol{d}^2\vec{r} = b_{ab}\,du^a du^b.} \qquad (6.53)$$

It is obvious that

$$\frac{\mathrm{II}}{ds^2} = \vec{n}\cdot\frac{\boldsymbol{d}^2\vec{r}}{ds^2} = \vec{n}\cdot\vec{a} = a_n \qquad (6.54)$$

is the *normal* component of the acceleration vector $\vec{a}$.

Gauss curvature. (??)

Gauss's formulas. When the point p move to another point q on $\mathcal{M}$, the frame also changes. The variations along u- and v-curves yield

$$\begin{cases} \boldsymbol{d}\vec{r}_u = \dfrac{\partial\vec{r}_u}{\partial u}\,du+\dfrac{\partial\vec{r}_u}{\partial v}\,dv, & (6.55a)\\ \boldsymbol{d}\vec{r}_v = \dfrac{\partial\vec{r}_v}{\partial u}\,du+\dfrac{\partial\vec{r}_v}{\partial v}\,dv. & (6.55b)\end{cases}$$

The partial derivatives of $\vec{r}_u$ and $\vec{r}_v$ with respect to u and v are new vectors, we can expand them by the frame $(p;\vec{r}_u,\vec{r}_u,\vec{n})$:

$$\vec{r}_{uu} := \frac{\partial\vec{r}_u}{\partial u} = \vec{r}_u\,(\Gamma_u)^u{}_u+\vec{r}_v\,(\Gamma_u)^v{}_u+\vec{n}\,(\Gamma_u)^n{}_u$$

$$= \vec{r}_u\,(\Gamma_u)^u{}_u + \vec{r}_v\,(\Gamma_u)^v{}_u + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{uu})}_{L}, \tag{6.56a}$$

$$\begin{aligned}\vec{r}_{uv} := \frac{\partial \vec{r}_u}{\partial v} &= \vec{r}_u\,(\Gamma_u)^u{}_v + \vec{r}_v\,(\Gamma_u)^v{}_v + \vec{n}\,(\Gamma_u)^n{}_v \\ &= \vec{r}_u\,(\Gamma_u)^u{}_v + \vec{r}_v\,(\Gamma_u)^v{}_v + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{uv})}_{M},\end{aligned} \tag{6.56b}$$

$$\begin{aligned}\vec{r}_{vu} := \frac{\partial \vec{r}_v}{\partial u} &= \vec{r}_u\,(\Gamma_v)^u{}_u + \vec{r}_v\,(\Gamma_v)^v{}_u + \vec{n}\,(\Gamma_v)^n{}_u \\ &= \vec{r}_u\,(\Gamma_v)^u{}_u + \vec{r}_v\,(\Gamma_v)^v{}_u + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{vu})}_{M},\end{aligned} \tag{6.56c}$$

$$\begin{aligned}\vec{r}_{vv} := \frac{\partial \vec{r}_v}{\partial v} &= \vec{r}_u\,(\Gamma_v)^u{}_v + \vec{r}_v\,(\Gamma_v)^v{}_v + \vec{n}\,(\Gamma_v)^n{}_v \\ &= \vec{r}_u\,(\Gamma_v)^u{}_v + \vec{r}_v\,(\Gamma_v)^v{}_v + \vec{n}\,\underbrace{(\vec{n}\cdot\vec{r}_{vv})}_{N}.\end{aligned} \tag{6.56d}$$

We call these set of equations the *Gauss's formulas*. We identify the coefficients

$$\boxed{(\Gamma_a)^c{}_b \equiv \Gamma^c{}_{ab},} \tag{6.57}$$

e.g., $(\Gamma_n)^u{}_v = \Gamma^u{}_{nv}$, which is called the *affine connection coefficients* (or simply the *affinity*), and we can particularly defined a tensor called *torsion tensor* as the antisymmetric part of the affinity

$$\boxed{T^c{}_{ba} := 2\,\Gamma^c{}_{[ab]} = \Gamma^c{}_{ab} - \Gamma^c{}_{ba}.} \tag{6.58}$$

Then Gauss formulas (6.56) now can be written as

$$\boxed{\vec{r}_{ab} = \vec{r}_c\,\Gamma^c{}_{ab} + \vec{n}\,\underbrace{\Gamma^n{}_{ab}}_{b_{ab}} := \vec{r}_c\,\Gamma^c{}_{ab} + \vec{n}\,b_{ab} \quad (a,b,c=u,v),} \tag{6.59}$$

and the displacement vector is

$$\boxed{\boldsymbol{d}\vec{r}_a = \vec{r}_c\,\Gamma^c{}_{ab}\,du^b + \vec{n}\,\Gamma^n{}_{ab}\,du^b := \vec{r}_c\,\Gamma^c{}_a + \vec{n}\,\Gamma^n{}_a} \tag{6.60}$$

where the coefficients are obtained by

$$\boxed{\Gamma_{cab} := \underbrace{g_{cd}}_{\vec{r}_c\cdot\vec{r}_d}\,\Gamma^d{}_{ab} = \vec{r}_c\cdot(\underbrace{\vec{r}_d\,\Gamma^d{}_{ab}}_{\vec{r}_{ab}-\vec{n}\,b_{ab}}) = \vec{r}_c\cdot\vec{r}_{ab}} \tag{6.61}$$

and

$$\boxed{b_{ab} = \Gamma^n{}_{ab} = \vec{n}\cdot\vec{r}_{ab}.} \tag{6.62}$$

We note that Γ_{cab} and b_{ab} are *symmetric* in indices a, b.

Weingarten's formulas. Now we can also consider the partial derivative of $\vec{n}$ with respect to u and v. However, due to $\vec{n}\cdot\vec{n}=1$, we have

$$\vec{n}_a\cdot\vec{n}=0 \quad\Longrightarrow\quad \vec{n}_a\perp\vec{n}, \tag{6.63}$$

i.e., $\vec{n}_u,\vec{n}_v\perp\vec{n}$ and $\{\vec{n}_a\}$ has no component along $\vec{n}$. So we have

$$\begin{cases}\vec{n}_u=\vec{r}_u\,(\Gamma_n)^u{}_u+\vec{r}_v\,(\Gamma_n)^v{}_u\equiv\vec{r}_u\,A+\vec{r}_v\,B, & (6.64a)\\ \vec{n}_v=\vec{r}_u\,(\Gamma_n)^u{}_v+\vec{r}_v\,(\Gamma_n)^v{}_v\equiv\vec{r}_u\,C+\vec{r}_v\,D, & (6.64b)\end{cases}$$

called the *Weingarten's formulas*. We calculate the inner product of $\vec{n}_u\cdot\vec{r}_u$ and $\vec{n}_u\cdot\vec{r}_v$:

$$\begin{cases}\vec{n}_u\cdot\vec{r}_u=-L=EA+FB\\ \vec{n}_u\cdot\vec{r}_v=-M=FA+GB\end{cases}\quad\Longrightarrow\quad \boxed{A=\frac{FM-GL}{EG-F^2}}\quad\text{and}\quad\boxed{B=\frac{FL-EM}{EG-F^2}.} \tag{6.65}$$

Similarly, by computing $\vec{n}_v\cdot\vec{r}_a$, we have

$$\boxed{C=\frac{FN-GM}{EG-F^2}}\quad\text{and}\quad\boxed{D=\frac{FM-EN}{EG-F^2}.} \tag{6.66}$$

Generally the Weingarten formulas (6.64) can be written by

$$\boxed{\vec{n}_b=\vec{r}_c\,\Gamma^c{}_{nb},} \tag{6.67}$$

where the coefficients can be calculated as follows

$$\underbrace{\vec{r}_a\cdot\vec{n}_b}_{-b_{ab}}=\vec{r}_a\cdot\left(\vec{r}_c\,\Gamma^c{}_{nb}\right)=g_{ac}\,\Gamma^c{}_{nb}:=\Gamma_{anb}\quad\Longrightarrow\quad\boxed{\Gamma^c{}_{nb}=g^{ca}\Gamma_{anb}=-g^{ca}b_{ab}:=-b^c{}_b.} \tag{6.68}$$

As a result, Weingarten formulas are read as

$$\boxed{\vec{n}_b=-\vec{r}_c\,b^c{}_b.} \tag{6.69}$$

6.4 Geodesic Equations

Consider a curve $\vec{r}(s)$ represented a trajectory of an object, which is parameterized by its arc length s on $\mathcal{M}$. The corresponding velocity vector (the tangent vector) at any point on the trajectory is

$$\vec{r}\,'(s):=\frac{\boldsymbol{d}\vec{r}(s)}{ds}=\frac{\partial\vec{r}}{\partial u^a}\frac{du^a}{ds}:=\vec{r}_a\,u^{a\prime}, \tag{6.70}$$

and it leads to the acceleration vector is given by

$$\begin{aligned}\vec{a}=\vec{r}\,''&=\vec{r}_a'\,u^{a\prime}+\vec{r}_a\,u^{a\prime\prime}\\ &=\left(\vec{r}_c\,\Gamma^c{}_{ab}+\vec{n}\,b_{ab}\right)u^{b\prime}u^{a\prime}+\vec{r}_a u^{a\prime\prime}\\ &\overset{a\leftrightarrow c}{=}\underbrace{\vec{r}_a\left(u^{a\prime\prime}+\Gamma^a{}_{cb}\,u^{c\prime}u^{b\prime}\right)}_{\text{tangential}}+\underbrace{\vec{n}\,b_{ab}\,u^{a\prime}u^{b\prime}}_{\text{normal}} && (6.71a)\\ &\equiv\vec{a}_t+\vec{a}_n && (6.71b)\end{aligned}$$

$$:= \vec{\kappa}_g + \vec{\kappa}_n, \tag{6.71c}$$

where we have used the chain rule and the Gauss's formulas to have

$$\vec{r}\,'_a = \vec{r}_{a\color{red}b}\,\frac{du^{\color{red}b}}{ds} = \big(\vec{r}_c\,\Gamma^c{}_{a\color{red}b} + \vec{n}\,b_{a\color{red}b}\big)u^{\color{red}b\,\prime}. \tag{6.72}$$

Now we call the curve $\vec{r}(s)$ *geodesic* if the curve is a *straight line* with $\vec{r}\,'' = \vec{a}_n =: \vec{\kappa}_n$ (the *normal curvature vector*), i.e., the *tangential* part of the acceleration vector *vanishes*

$$\boxed{\vec{a}_t = \vec{r}_a\big(u^{a\,\prime\prime} + \Gamma^a{}_{cb}\,u^{c\,\prime}u^{b\,\prime}\big) = 0,} \tag{6.73}$$

which is identical to the following statement:

> A curve $\vec{r}(s)$ *only* has the normal curvature $\vec{\kappa}_n$ and $\vec{r}\,''$ *only* changes the direction of a moving object.

That means the *tangential* velocity $\vec{V}_t = (\vec{r}\,')_t$ of the object is *constant* and the motion on $\mathcal{M}$ satisfies the law of inertia of Galileo. Thus the object only feels a force *vertically*. Because $\vec{r}_a$s are linearly independent, (6.73) is reduced to

$$\boxed{u^{a\,\prime\prime} + \Gamma^a{}_{cb}\,u^{c\,\prime}u^{b\,\prime} = 0,} \tag{6.74}$$

which is called the *geodesic equation* and the curve $\vec{r}(s)$ or its trajectory $u^a(s)$ is always called the *geodesic* (*curve*), and, therefore,

$$\boxed{\vec{\kappa}_g := \vec{a}_t} \tag{6.75}$$

is called the *geodesic curvature vector*.

Covariant derivative. Generally, given an arbitrary vector $\vec{X} = \vec{r}_a\,X^a$, we have

$$\begin{aligned}\frac{\boldsymbol{d}\vec{X}}{ds} = \vec{X}\,' &= \vec{r}\,'_a\,X^a + \vec{r}_a\,X^{a\,\prime} \\ &= \underbrace{\vec{r}_a\big(X^{a\,\prime} + \Gamma^a{}_{cb}\,X^c u^{b\,\prime}\big)}_{\text{tangential}} + \underbrace{\vec{n}\,b_{a\color{red}b}\,X^a u^{\color{red}b\,\prime}}_{\text{normal}}.\end{aligned} \tag{6.76}$$

We can use the chain rule to rewrite $X^{a\,\prime}$ by

$$X^{a\,\prime} = \frac{\partial X^a}{\partial u^{\color{red}b}}\frac{du^{\color{red}b}}{ds} = X^a{}_{,\color{red}b}\,u^{\color{red}b\,\prime} \tag{6.77}$$

as a product of a partial derivative with respect to the coordinates $u^{\color{red}b}$ and $u^{\color{red}b\,\prime}$ by chain rule. Substituting the expression into (6.76), then $\vec{X}\,'$ would be read by

$$\vec{X}\,' = \underbrace{\vec{r}_a\big(X^a{}_{,\color{red}b} + \Gamma^a{}_{c\color{red}b}\,X^c\big)u^{\color{red}b\,\prime}}_{(\vec{X}\,')_t := (\vec{X}_{\color{red}b})_t\,u^{\color{red}b\,\prime}} + \underbrace{\vec{n}\,b_{a\color{red}b}\,X^a u^{\color{red}b\,\prime}}_{(\vec{X}\,')_n := (\vec{X}_{\color{red}b})_n\,u^{\color{red}b\,\prime}} := \vec{X}_{\color{red}b}\,u^{\color{red}b\,\prime}, \tag{6.78}$$

where t and n denote the *tangential* and *normal* parts respectively. We define the *tangential* part of the derivative of $\vec{X}$ by a new differentiation $\boldsymbol{D}$:

$$\boxed{\left(\vec{X}'\right)_t := \frac{\boldsymbol{D}\vec{X}}{ds} := \boldsymbol{D}_{\vec{r}\,'}\vec{X} := u^{b\,\prime}\left(\boldsymbol{D}_b\vec{X}\right)} \tag{6.79a}$$

with

$$\boxed{\frac{\boldsymbol{D}}{ds} := \boldsymbol{D}_{\vec{r}\,'} = \boldsymbol{D}_{(\vec{r}_b u^{b\,\prime})} = u^{b\,\prime}\boldsymbol{D}_{\vec{r}_b} \equiv u^{b\,\prime}\boldsymbol{D}_b\,,} \tag{6.79b}$$

which is the directional derivative along the direction of the tangent vector $\vec{r}\,' = \vec{r}_b\, u^{b\,\prime}$ of a trajectory $u^b(s)$ on $\mathcal{M}$ and where

$$\boldsymbol{D}_b\vec{X} \equiv \vec{r}_a\left(\boldsymbol{D}_b\vec{X}\right)^a := \left(\vec{X}_b\right)_t = \vec{r}_a\left(X^a{}_{,b} + \Gamma^a{}_{cb}\,X^c\right), \tag{6.80}$$

is the so-called *covariant derivative* of $\vec{X}$ with respect to the coordinate u^b as the *tangential part* of $\vec{X}_{,b}$ on $\mathcal{M}$ by using the *bold* symbol $\boldsymbol{D}$. We note that the action of the *covariant derivative operator* $\boldsymbol{D}_b$ on the vector $\vec{X}$ is

$$\boldsymbol{D}_b\vec{X} = \boldsymbol{D}_b\left(\vec{r}_a\,X^a\right) = \left(\boldsymbol{D}_b\vec{r}_a\right)X^a + \vec{r}_a\left(\boldsymbol{D}_b X^a\right). \tag{6.81}$$

Hence, by comparing (6.80) and (6.81), the action of $\boldsymbol{D}_b$ on basis of $\vec{r}_a$ and component X^a respectively are

$$\boxed{\begin{cases} \boldsymbol{D}_b\vec{r}_a = (\vec{r}_{ab})_t = \vec{r}_c\,\Gamma^c{}_{ab}\,, & \text{(6.82a)} \\ \boldsymbol{D}_b X^a = \partial_b X^a \overset{\text{or}}{=} X^a{}_{,b}\,. & \text{(6.82b)} \end{cases}}$$

Alternatively we can define the covariant derivative in another form denoted by ∇ with different differential properties, which is always used by physicists to work on the component computations in the *tensor calculus*:

$$\boxed{\begin{cases} \nabla_b X^a \equiv X^a{}_{;b} := \left(\boldsymbol{D}_b\vec{X}\right)^a = X^a{}_{,b} + \Gamma^a{}_{cb}\,X^c, & \text{(6.83a)} \\ \nabla_b\vec{r}_a \equiv \vec{r}_{a;b} = 0. & \text{(6.83b)} \end{cases}}$$

As a result, $\nabla_b X^a$ is the *component expression* of $\boldsymbol{D}_b\vec{X}$ in the direction of $\vec{r}_a$. Therefore, we note that ∇ has nothing to do on the basis vectors. The equivalence of two operators can be clarified as follows:

$$\boldsymbol{D}_b\vec{X} = \vec{r}_a(\boldsymbol{D}_b\vec{X})^a = \vec{r}_a(\underbrace{\nabla_b X^a}_{X^a{}_{;b}}) + \overbrace{0}^{0\cdot X^a} := \nabla_b\vec{X} \overset{\text{or}}{=} \vec{X}_{;b} \tag{6.84}$$

$$\boxed{0 = \nabla_b\vec{r}_a \overset{\text{or}}{=} \vec{r}_{a;b}}$$

such that

$$\boxed{\frac{\nabla\vec{X}}{ds} = \vec{r}_a\frac{\nabla X^a}{ds} := \vec{r}_a(\nabla_{\vec{r}\,'}X^a) = \vec{r}_a(u^{b\,\prime}\,\nabla_b X^a) \overset{\text{or}}{=} \vec{r}_a\left(X^a{}_{;b}\,u^{b\,\prime}\right).} \tag{6.85}$$

Parallel transport. We will say that any vector $\vec{X}$ is parallel to the curve $\vec{r}(s)$ if its *tangential* part of the derivative vanishes

$$\left(\vec{X}'\right)_t = \frac{\boldsymbol{D}\vec{X}}{ds} = \frac{\nabla\vec{X}}{ds} = \vec{r}_a\frac{\nabla X^a}{ds} = \vec{r}_a\left(u^{b\prime}\nabla_b X^a\right) = 0 \quad \Longrightarrow \quad \boxed{u^{b\prime}\nabla_b X^a = 0.} \tag{6.86}$$

Since $u^{b\prime}$ is arbitrary but non-zero, the so-called *parallel transport* of a vector X^a can also be written as

$$\boxed{\nabla_b X^a = \partial_b X^a + \Gamma^a{}_{cb}X^c = 0} \quad \text{with the operator} \quad \boxed{\left(\nabla_b\right)^a{}_c = \delta^a_c\partial_b + \Gamma^a{}_{cb}} \tag{6.87a}$$

or

$$\boxed{\partial_b X^a = -\Gamma^a{}_{cb}X^c.} \tag{6.87b}$$

Parallel transport of a vector can be realized as follows. Let a curve $\vec{r}(s)$ parameterized by s. If

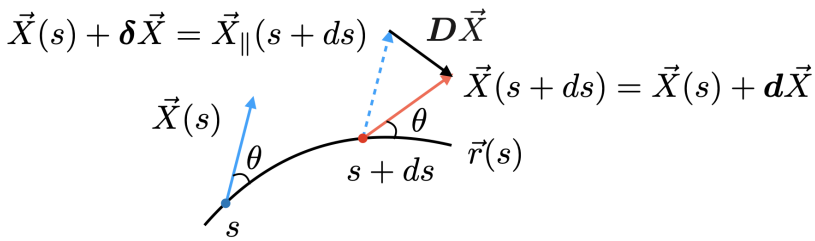


Fig. 6.7: Parallel transport of vector $\vec{X}$.

we are interested in a vector $\vec{X}(s)$ parallel transported from s to $s+ds$, we have to look for the relation between vectors $\vec{X}(s)$ and $\vec{X}(s+ds)$. Let the transported vector at $s+ds$ defined by

$$\vec{X}_{\parallel}(s+ds) = \vec{X}(s) + \boldsymbol{\delta}\vec{X}, \tag{6.88}$$

where $\boldsymbol{\delta}\vec{X}$ should be linear in the change of coordinate and the vector itself, then it is given by

$$\boxed{\boldsymbol{\delta}\vec{X} = -\vec{r}_a\,\underbrace{\Gamma^a{}_{cb}\,X^c\,du^b}_{\delta X^c}.} \tag{6.89}$$

By the way, the vector defined at $s+ds$ can be expanded as

$$\vec{X}(s+ds) = \vec{X}(s) + \boldsymbol{d}\vec{X} \quad \text{with} \quad \boxed{\boldsymbol{d}\vec{X} = \vec{X}'\,ds = \vec{r}_a\,\underbrace{X^a{}_{,b}\,du^b}_{dX^a}.} \tag{6.90}$$

Then we define $\boldsymbol{D}\vec{X}$ by

$$\boxed{\begin{aligned}\boldsymbol{D}\vec{X} &\coloneqq \vec{X}(s+ds) - \vec{X}_{\parallel}(s+ds)\\ &= \boldsymbol{d}\vec{X} - \boldsymbol{\delta}\vec{X},\end{aligned}} \tag{6.91}$$

which is illustrated in Fig. 6.7. If $\vec{X}(s)$ is parallel transported from s to $s+ds$, it means that the angle θ between $\vec{X}$ and $\vec{r}(s)$ is preserved and the transported vector should be equal to the vector defined at the transported point at $s+ds$, i.e.,

$$\boxed{\vec{X}(s+ds) = \vec{X}_{\parallel}(s+ds)} \quad \Longrightarrow \quad \boxed{\boldsymbol{D}\vec{X} = 0} \quad \text{or} \quad \boxed{\boldsymbol{d}\vec{X} = \boldsymbol{\delta}\vec{X}.} \tag{6.92}$$

So we deserves the same equation as (6.87a). Particularly, we call a vector $\vec{X}$ *uniform* if $\vec{X}$ satisfies the condition of the parallel transport $\boldsymbol{D}\vec{X} = 0$.

Furthermore, if a vector $\vec{X} = \vec{r}\,'(s)\,\vec{r}_a\,u^{a\,\prime} := \vec{r}_a V^a = \vec{V}$ is the tangent vector of a geodesic, the component of the equation of the parallel transport of a vector $\vec{X} = \vec{V}$ is

$$\left(\vec{V}'\right)^a_t = \left(\frac{\boldsymbol{D}\vec{V}}{ds}\right)^a = \frac{\nabla V^a}{ds} = V^b\left(\nabla_b V^a\right) = 0, \tag{6.93}$$

and by the definition of the covariant derivative, we found

$$u^{a\,\prime\prime} + \Gamma^a{}_{cb}\,u^{c\,\prime}u^{b\,\prime} = u^{b\,\prime}\frac{\partial u^{a\,\prime}}{\partial u^b} + \Gamma^a{}_{cb}\,u^{c\,\prime}u^{b\,\prime} = V^a{}_{,b}\,V^b + \Gamma^a{}_{cb}\,V^c V^b = V^b\nabla_b V^a, \tag{6.94}$$

thus the geodesic equations (6.74) can be rewritten as

$$\boxed{V^b\nabla_b V^a \equiv V^a{}_{;b}\,V^b = 0,} \tag{6.95}$$

which coincides with the equation of parallel transport.

Note: As we have mentioned that we will always concern a n-dimensional space $\mathcal{M}$ which is not a subspace of other space, so we have no second fundamental form for $\mathcal{M}$, i.e., $b_{ij} = 0$, and $\vec{V}' = \left(\vec{V}'\right)_t$ is tangent to $\mathcal{M}$, indeed. Hence, the derivative of a tangent vector should be

$$\begin{aligned}\vec{V}' &= \frac{\boldsymbol{D}\vec{V}}{ds} = V^i\,\boldsymbol{D}_i\vec{V} \\ &\overset{\text{or}}{=} e_j\,\frac{\nabla V^j}{ds} = e_j\left(V^i\,\nabla_i V^j\right) \quad (i,j = 1,2,\cdots,n).\end{aligned} \tag{6.96}$$

Covariant derivative of a metric tensor. We assume that the magnitude of $\vec{X}$ is preserved under the parallel transport, additionally the norm $\|\vec{X}\|^2$ is a scalar, we would have

$$\begin{aligned}0 &= \frac{d}{ds}\|\vec{X}\|^2 = \frac{\nabla}{ds}\|\vec{X}\|^2 \\ &= \left(u^{c\,\prime}\,\nabla_c\right)\left(g_{ab}X^aX^b\right) \\ &= \left(u^{c\,\prime}\,\nabla_c\,g_{ab}\right)X^aX^b + g_{ab}\big(\underbrace{u^{c\,\prime}\,\nabla_c X^a}_{0}\big)X^b + g_{ab}X^a\big(\underbrace{u^{c\,\prime}\,\nabla_c X^b}_{0}\big).\end{aligned} \tag{6.97}$$

The result gives the covariant derivative of a metric tensor

$$\boxed{\nabla_c\,g_{ab} = 0,} \tag{6.98}$$

which is the well-known *metric compatible condition* or *metricity condition*. It implies that the metric of a space is preserved under the parallel transport. It can be shown that the *cosine* between $\vec{X} = \vec{r}_a X^a$ and tangent vector $\vec{r}\,' = \vec{r}_b\,u^{b\,\prime}$, which stands for by $\vec{r}_a\cdot\vec{r}_b = \|\vec{r}_a\|\|\vec{r}_b\|\cos\theta = g_{ab}$, stays the *same* under the parallel transport:

$$\frac{\nabla}{ds}\left(\vec{r}_a\cdot\vec{r}_b\right) = \frac{\nabla}{ds}\left(\|\vec{r}_a\|\|\vec{r}_b\|\cos\theta\right) = \underbrace{\frac{d\|\vec{r}_a\|}{ds}}_{0}\,\|\vec{r}_b\|\,\cos\theta + \|\vec{r}_a\|\,\underbrace{\frac{d\|\vec{r}_b\|}{ds}}_{0}\,\cos\theta + \|\vec{r}_a\|\|\vec{r}_a\|\,\frac{d\cos\theta}{ds} \tag{6.99}$$

However,

$$\frac{\nabla}{ds}\left(\vec{r}_a \cdot \vec{r}_b\right) = \left(u^{c\prime}\,\nabla_c\right) g_{ab} = u^{c\prime}\,\underbrace{\nabla_c\, g_{ab}}_{0} = 0. \tag{6.100}$$

So, we would have

$$\frac{d\cos\theta}{ds} = 0 \quad \Longrightarrow \quad \boxed{\theta = \text{const.}} \tag{6.101}$$

This consequence can be illustrated by Fig. 6.7 through $\boldsymbol{D}\vec{X} = 0$.

On the other hand, we can do the derivative of $\|\vec{X}\|^2$ with respect to s directly:

$$\begin{aligned}
\frac{d}{ds}\|\vec{X}\|^2 &= \frac{d}{ds}\left(g_{ab}X^aX^b\right) \\
&= \frac{1}{ds}\Big[\left(dg_{ab}\right)X^aX^b + g_{ab}\left(dX^aX^b\right) + g_{ab}X^a\left(dX^b\right)\Big] \\
&= \frac{1}{ds}\Big[\left(du^c\,\partial_c\, g_{ab}\right)X^aX^b + g_{ab}\big(\underbrace{dX^a}_{\delta X^a = -\Gamma^a{}_{dc}X^d\,du^c}\big)X^b + g_{ab}X^a\big(\underbrace{dX^b}_{\delta X^b = -\Gamma^b{}_{dc}X^d\,du^c}\big)\Big] \\
&= \left(\frac{du^c}{ds}\,\partial_c\, g_{ab}\right)X^aX^b - g_{ab}\left(\Gamma^a{}_{dc}X^d\,\frac{du^c}{ds}\right)X^b - g_{ab}X^a\left(\Gamma^b{}_{dc}X^d\,\frac{du^c}{ds}\right) \\
&= u^{c\prime}\left(\partial_c\, g_{ab} - g_{db}\Gamma^d{}_{ac} - g_{ad}\Gamma^d{}_{bc}\right)X^aX^b.
\end{aligned} \tag{6.102}$$

As a result, we can equal two results and obtain the rule for calculating the covariant derivative of a metric tensor

$$\boxed{\begin{aligned}\nabla_c\, g_{ab} &= \partial_c\, g_{ab} - \Gamma^d{}_{ac}\, g_{db} - \Gamma^d{}_{bc}\, g_{ad} \\ &= \partial_c\, g_{ab} - \Gamma_{bac} - \Gamma_{abc}.\end{aligned}} \tag{6.103}$$

It can be shown that the covariant derivative of g^{ab} should be

$$\boxed{\nabla_c\, g^{ab} = \partial_c\, g^{ab} + \Gamma^a{}_{dc}\, g^{db} + \Gamma^b{}_{dc}\, g^{ad}.} \tag{6.104}$$

Covariant derivative of an arbitrary rank tensor. For an arbitrary rank tensor, for example a (r, s)-type tensor, its covariant derivative can be generalized from the result of the metric tensor:

$$\boxed{\begin{aligned}\nabla_c\, K^{a_1a_2\cdots a_r}{}_{b_1b_2\cdots b_s} &= \partial_c\, K^{a_1a_2\cdots a_r}{}_{b_1b_2\cdots b_s} \\ &\quad + \Gamma^{a_1}{}_{dc}\, K^{da_2\cdots a_r}{}_{b_1b_2\cdots b_s} + \Gamma^{a_2}{}_{dc}\, K^{a_1d\cdots a_r}{}_{b_1b_2\cdots b_s} \\ &\quad + \cdots + \Gamma^{a_r}{}_{dc}\, K^{a_1a_2\cdots d}{}_{b_1b_2\cdots b_s} \\ &\quad - \Gamma^d{}_{b_1c}\, K^{a_1a_2\cdots a_r}{}_{db_2\cdots b_s} - \Gamma^d{}_{b_2c}\, K^{a_1a_2\cdots a_r}{}_{b_1d\cdots b_s} \\ &\quad - \cdots - \Gamma^d{}_{b_sc}\, K^{a_1a_2\cdots a_r}{}_{b_1b_2\cdots d}.\end{aligned}} \tag{6.105}$$

<u>Note:</u> It is easy to obtain the result of $\nabla_c\, K^{a_1a_2\cdots a_r}{}_{b_1b_2\cdots b_s}$ by applying $\boldsymbol{D}$ to the tensor K:

$$\boxed{\boldsymbol{D}_cK = \left(\nabla_c\, K^{a_1a_2\cdots a_r}{}_{b_1b_2\cdots b_s}\right)\vec{r}_{a_1}\otimes\vec{r}_{a_2}\otimes\cdots\otimes\vec{r}_{a_r}\otimes du^{b_1}\otimes du^{b_2}\otimes\cdots\otimes du^{b_s},} \tag{6.106}$$

and $\{du^a\}$ is the dual basis vector of $\{\vec{r}_b\}$ such that

$$\boxed{du^a(\vec{r}_b) = \delta^a_b} \quad \text{and} \quad \boxed{\boldsymbol{D}_c\, du^a = -\Gamma^a{}_{bc}\, du^b.} \tag{6.107}$$

Tensor density. Given a fundamental completely antisymmetric tensor, the Levi–Civita tensor,

$$\boldsymbol{\eta}_{abc} \coloneqq \sqrt{\mathbf{g}}\, \varepsilon_{abc}. \tag{6.108}$$

Then we apply the covariant derivative on the component of this tensor of $\boldsymbol{\eta}_{123}$:

$$\begin{aligned}
\nabla_a \boldsymbol{\eta}_{123} &= \partial_a\big(\sqrt{\mathbf{g}}\, \varepsilon_{123}\big) + \sqrt{\mathbf{g}}\,\big(-\Gamma^b{}_{1a} \underbrace{\varepsilon_{b23}}_{b=1} - \Gamma^b{}_{2a} \underbrace{\varepsilon_{1b3}}_{b=2} - \Gamma^b{}_{3a} \underbrace{\varepsilon_{12b}}_{b=3}\big) \\
&= \big(\partial_a \sqrt{\mathbf{g}}\big)\varepsilon_{123}\big) - \sqrt{\mathbf{g}}\,\big(\Gamma^1{}_{1a}\, \varepsilon_{123} + \Gamma^2{}_{2a}\, \varepsilon_{123} + \Gamma^3{}_{3a}\, \varepsilon_{123}\big) \\
&= \big(\partial_a \sqrt{\mathbf{g}}\big)\varepsilon_{123} - \sqrt{\mathbf{g}}\, \Gamma^c{}_{ca}\, \varepsilon_{123} \\
&= \big(\partial_a \sqrt{\mathbf{g}} - \Gamma^c{}_{ca}\, \sqrt{\mathbf{g}}\big)\, \varepsilon_{123}.
\end{aligned} \tag{6.109}$$

However, the covariant derivative can be written by

$$\nabla_a \boldsymbol{\eta}_{123} = \big(\nabla_a \sqrt{\mathbf{g}}\big)\varepsilon_{123}. \tag{6.110}$$

By comparing (6.109) and (6.110), we obtain the formula of

$$\boxed{\nabla_a \sqrt{\mathbf{g}} = \partial_a \sqrt{\mathbf{g}} - \Gamma^c{}_{ca}\, \sqrt{\mathbf{g}}\,.} \tag{6.111}$$

Now we write a tensor density $\mathfrak{A}^{\cdots}{}_{\cdots}$ associated to the corresponding tensor $A^{\cdots}{}_{\cdots}$ by

$$\mathfrak{A}^{\cdots}{}_{\cdots} = \sqrt{\mathbf{g}}\, A^{\cdots}{}_{\cdots}\,. \tag{6.112}$$

Covariant derivative of $\mathfrak{A}^{\cdots}{}_{\cdots}$

6.5 Christoffel Symbol and Curvature Tensor

According to (6.61), we can obtain the following equations

$$\begin{cases}
\dfrac{\partial}{\partial u^c}\, g_{ab} = \dfrac{\partial}{\partial u^c}(\vec{r}_a \cdot \vec{r}_b) = \vec{r}_{ac} \cdot \vec{r}_b + \vec{r}_a \cdot \vec{r}_{bc} = \Gamma_{bac} + \Gamma_{abc}\,, & \text{(6.113a)} \\
\dfrac{\partial}{\partial u^a}\, g_{bc} = \Gamma_{cba} + \Gamma_{bca}\,, & \text{(6.113b)} \\
\dfrac{\partial}{\partial u^b}\, g_{ca} = \Gamma_{acb} + \Gamma_{cab}\,. & \text{(6.113c)}
\end{cases}$$

Equations above are, in fact, the metricity condition of the covariant derivative of g_{ab}. Through the combination (6.113c) + (6.113b) − (6.113a), with the help of symmetry

$$\vec{r}_{ab} = \frac{\partial^2 \vec{r}}{\partial u^a \partial u^b} = \vec{r}_{ba}, \tag{6.114}$$

the specific solution of the affinity Γ_{cab} is a function of the metric tensor g_{ab}:

$$\boxed{\Gamma_{cab} = [c, ab],} \tag{6.115}$$

where the shorthand notation

$$\boxed{[c, ab] \coloneqq \frac{1}{2}\left(\frac{\partial}{\partial u^a}\, g_{bc} + \frac{\partial}{\partial u^b}\, g_{ca} - \frac{\partial}{\partial u^c}\, g_{ab}\right)} \tag{6.116}$$

is called the *Christoffel symbol of the first kind.*[†14] For the first-index raised affinity $\Gamma^d{}_{ab} = g^{dc}\Gamma_{cab}$, we have

$$\boxed{\Gamma^d{}_{ab} = g^{dc}[c, ab] \coloneqq \left\{{d \atop ab}\right\},} \tag{6.117}$$

where the shorthand is

$$\boxed{\left\{{d \atop ab}\right\} \coloneqq \frac{1}{2} g^{dc}\left(\frac{\partial}{\partial u^a}\, g_{bc} + \frac{\partial}{\partial u^b}\, g_{ca} - \frac{\partial}{\partial u^c}\, g_{ab}\right)} \tag{6.118}$$

called the *Christoffel symbol of the second kind.*[†15] It is obvious that we have the particular *symmetry* of the solution Γ_{cab} in indices a and b, i.e.,

$$[c, ab] = \partial_{(a}\, g_{b)c} - \frac{1}{2}\,\partial_c\, g_{ab} = [c, ba] \quad \text{and} \quad T_{cba} = 2\,\Gamma_{c[ab]} = 0. \tag{6.119}$$

The solution $[c, ab]$ (or $\left\{{c \atop ab}\right\}$) is the well-known coefficients of the *Levi–Civita connection*, which indicates that the space considered here is *Riemannian*. Here I would like to remind you that the solution $[c, ab]$ we obtained is implicitly using the conditionds of (i) metricity condition in (6.113), i.e., $\nabla_c\, g_{ab} = 0$; and (ii) the symmetry of $\vec{r}_{ab} = \vec{r}_{ba}$ in (6.114) (this implies $\Gamma_{cab} = \Gamma_{cba}$, which is called the *torsion-free condition*).

Before continuing the discussion of Christoffel symbol, we need to remind you two points of a matrix $\mathsf{A} \coloneqq (A_{ij})$:

(i) The inverse of a $n \times n$ matrix A is

$$\boxed{(\mathsf{A}^{-1}) \coloneqq \frac{\mathrm{adj}(\mathsf{A})}{\mathbf{A}}} \quad \text{with} \quad \boxed{\mathrm{adj}(\mathsf{A}) \coloneqq \mathrm{cof}(\mathsf{A})^T,} \tag{6.120}$$

i.e.,

$$\mathrm{adj}(\mathsf{A})_{ji} \coloneqq \Big(\mathrm{cof}(\mathsf{A})^T\Big)_{ji} = \mathrm{cof}(\mathsf{A})_{ij}. \tag{6.121}$$

Here $\mathbf{A} \coloneqq \det(A_{ij})$ is the determinant of A, $\mathrm{adj}(\mathsf{A})$ and $\mathrm{cof}(\mathsf{A})$ are called *adjoint matrix* and *cofactor matrix* of A respectively. The element in the cofactor matrix of

[†14] Some people may prefer the notation by $[ab, c]$. In Einstein's paper, you will find that he use the notation $\left[{ab \atop c}\right]$ as the Christoffel symbol of the first kind.

[†15] Einstein use the notation of $\left\{{ab \atop c}\right\}$ to denote the Christoffel symbol of the second kind.

A is defined by

$$\mathrm{cof}(\mathsf{A})_{ij} \equiv C_{ij}^{(\mathsf{A})} := (-1)^{i+j}\, M_{ij}^{(\mathsf{A})}. \tag{6.122}$$

Here $M_{ij}^{(\mathsf{A})}$ is called the *minor* of A_{ij}, which is defined by the determinant of the $(n-1)\times(n-1)$ matrix by eliminating the ith-row and jth-column of A:

$$M_{ij}^{(\mathsf{A})} := \det \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1j} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2j} & \cdots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ A_{i1} & A_{i2} & \cdots & A_{ij} & \cdots & A_{in} \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots \\ A_{n1} & A_{n2} & \cdots & A_{nj} & \cdots & A_{nn} \end{pmatrix}, \tag{6.123}$$

where the highlighted row and column above means that the ith-row and jth-column should be eliminated from the matrix A.

(ii) The differential of $\mathbf{A} := \det(A_{ij}) = \sum_i \mathrm{adj}(\mathsf{A})_{ji}\, A_{ij}$ for any index j, by using (6.120), is

$$\boxed{d\mathbf{A} = \sum_i \mathrm{adj}(\mathsf{A})_{ji}\, dA_{ij} = \sum_i \mathbf{A}(\mathsf{A}^{-1})_{ji}\, dA_{ij}.} \tag{6.124}$$

So we can apply (6.124) to the metric tensor g_{ab}, the differential (or derivative) of $\mathbf{g} := \det(g_{ab})$ would be

$$\boxed{d\mathbf{g} = \mathbf{g}\, g^{ac}\, dg_{ca}} \quad \text{or} \quad \boxed{\partial_b \mathbf{g} = \mathbf{g}\, g^{ac}\, \partial_b g_{ca} \overset{\text{or}}{=} \mathbf{g}\, g^{ac}\, g_{ca,\,b}} \tag{6.125}$$

and it leads to

$$d\sqrt{\mathbf{g}} = \frac{1}{2}\frac{d\mathbf{g}}{\sqrt{\mathbf{g}}} \quad \text{or} \quad \partial_b \sqrt{\mathbf{g}} = \frac{1}{2}\frac{\partial_b \mathbf{g}}{\sqrt{\mathbf{g}}}. \tag{6.126}$$

For a particular case of $c = a$, the Christoffel symbol is

$$\left\{{a \atop ab}\right\} = \frac{1}{2}\, g^{ac}\left(\cancel{\frac{\partial}{\partial u^a}\, g_{bc}} + \frac{\partial}{\partial u^b}\, g_{ca} - \cancel{\frac{\partial}{\partial u^c}\, g_{ab}}\right) = \boxed{\frac{1}{2}\, g^{ac}\, \partial_b g_{ca} \overset{\text{or}}{=} \frac{1}{2}\, g^{ac}\, g_{ca,\,b}\,.} \tag{6.127}$$

Therefore, by applying (6.125) and (6.126), (6.127) becomes

$$\boxed{\left\{{a \atop ab}\right\} = \underbrace{\frac{1}{2}\frac{\partial_b \mathbf{g}}{\mathbf{g}}}_{\frac{1}{2}\partial_b\left(\ln \mathbf{g}\right) \text{ or } \frac{1}{2}\left(\ln \mathbf{g}\right)_{,\,b}} = \frac{1}{2}\,\frac{2\sqrt{\mathbf{g}}\,\left(\partial_b \sqrt{\mathbf{g}}\right)}{\mathbf{g}} = \underbrace{\frac{\partial_b \sqrt{\mathbf{g}}}{\sqrt{\mathbf{g}}}}_{\partial_b\left(\ln \sqrt{\mathbf{g}}\right) \text{ or } \left(\ln \sqrt{\mathbf{g}}\right)_{,\,b}}} \quad \Longrightarrow \quad \boxed{\partial_b \sqrt{\mathbf{g}} \overset{\text{or}}{=} \sqrt{\mathbf{g}}_{\,,\,b} = \sqrt{\mathbf{g}}\, \left\{{a \atop ab}\right\}.} \tag{6.128}$$

Covariant divergence of a vector field. Given a vector V^a, its covariant derivative is

$$\nabla_b V^a = \partial_b V^a + \Gamma^a{}_{cb}\, V^c, \tag{6.129}$$

which leads to the *covariant divergence* of V^a

$$\boxed{\nabla_a V^a = \partial_a V^a + \underbrace{\Gamma^a{}_{ca}}_{2\,\Gamma^a{}_{[ca]}+\Gamma^a{}_{ac}} V^c = \partial_a V^a + \Gamma^a{}_{ac} V^c + \underbrace{2\,\Gamma^a{}_{[ca]}}_{T^a{}_{ac}} V^c.} \tag{6.130}$$

For particular case of $\Gamma^a{}_{bc} = \left\{{a \atop bc}\right\}$, since $\Gamma^a{}_{[ca]} = \left\{{a \atop [ca]}\right\} = 0$ and (6.128), the covariant divergence (6.130) would reduce to

$$\boxed{\nabla_a V^a = \partial_a V^a + \frac{\partial_c \sqrt{\mathbf{g}}}{\sqrt{\mathbf{g}}} V^c = \frac{1}{\sqrt{\mathbf{g}}}\,\partial_a\big(\sqrt{\mathbf{g}} V^a\big).} \tag{6.131}$$

We should note that (6.131) is a *special case* which is *only* valid in *Riemannian space* because in general $\Gamma^a{}_{bc}$ may *not* be the Levi–Civita cnonnection.

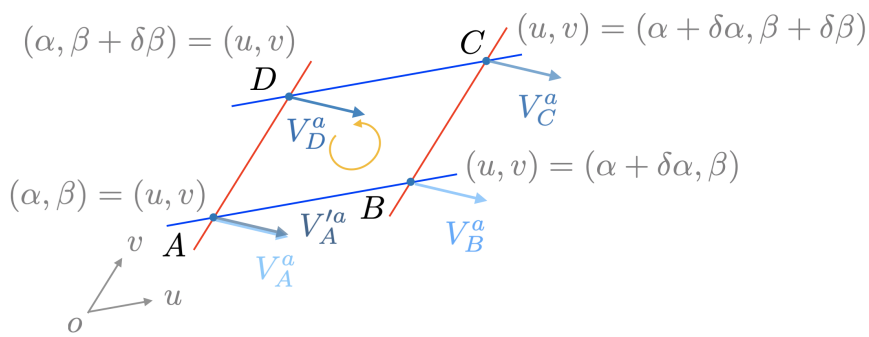


Fig. 6.8: Parallel transport of a vector V^a around a closed loop counterclockwise in a flat space.

Curvature tensor. Let's consider a situation that a vector V^a tangent to a *two-dimensional* surface $\mathcal{M}$ is parallel transported around a closed loop *counterclockwise* on $\mathcal{M}$. Cases in flat and curved space are respectively illustrated by Fig. 6.8 and Fig. 6.9. It is obvious that the vector V^a stays the same through the parallel transport if $\mathcal{M}$ is a flat space, i.e., $\mathcal{M} = \mathbb{R}^2$. However, it is different in the case of the curved space. Through the equation of parallel transport (6.87b) or (6.92), we would obtain

$$dV^a = \delta V^a = -\Gamma^a{}_{cu} V^c\, du \tag{6.132}$$

if the vector V^a is is parallel transported from A to B along the curve $v = \beta$. By integrating the equation along u from the point A to point B, the vector V^a at point B should be

$$\begin{aligned} A \to B: \quad V^a_B &= V^a_A + \int_A^B dV^a \\ &= V^a_A + \int_A^B du\, \frac{\partial V^a}{\partial u} \\ &= V^a_A - \int_A^B du\, \big(\Gamma^a{}_{cu} V^c\big)_{v=\beta} \\ &= V^a_A - \int_\alpha^{\alpha+\delta\alpha} du\, \big(\Gamma^a{}_{cu} V^c\big)_{v=\beta}. \end{aligned} \tag{6.133a}$$

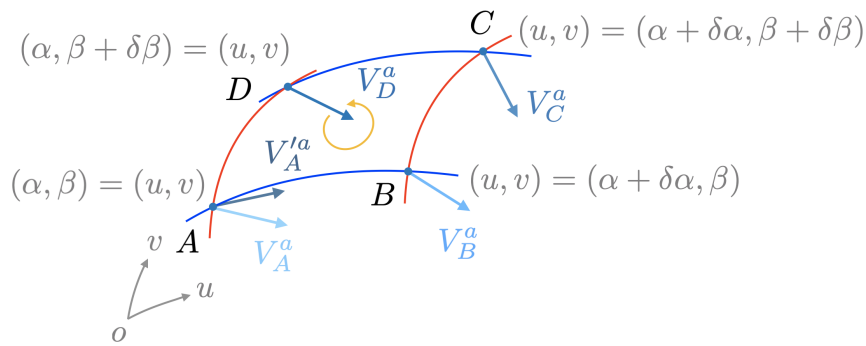


Fig. 6.9: Parallel transport of a vector V^a around a closed loop counterclockwise in a curved space.

Similarly, we have the integrations along v from the point B to point C, along u from the point C to point D and along v from the point D to point A

$$
\begin{aligned}
B \to C: \quad V_C^a &= V_B^a - \int_B^C dv \left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha+\delta\alpha} \\
&= V_B^a - \int_\beta^{\beta+\delta\beta} dv \left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha+\delta\alpha}, \qquad (6.133b)\\
C \to D: \quad V_D^a &= V_C^a - \int_C^D d(-u) \left(\Gamma^a{}_{cu} V^c\right)_{v=\beta+\delta\beta} \\
&= V_C^a + \int_\alpha^{\alpha+\delta\alpha} du \left(\Gamma^a{}_{cu} V^c\right)_{v=\beta+\delta\beta}, \qquad (6.133c)\\
D \to A: \quad V_A'^a &= V_D^a - \int_D^A d(-v) \left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha} \\
&= V_D^a + \int_\beta^{\beta+\delta\beta} dv \left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha}. \qquad (6.133d)
\end{aligned}
$$

As a result, the vector V^a at point A changes and the variation should be

$$
\begin{aligned}
\Delta V_A^a &:= V_A'^a - V_A^a \\
&= -\int_\alpha^{\alpha+\delta\alpha} du \left(\Gamma^a{}_{cu} V^c\right)_{v=\beta} - \int_\beta^{\beta+\delta\beta} dv \left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha+\delta\alpha} \\
&\quad + \int_\alpha^{\alpha+\delta\alpha} du \left(\Gamma^a{}_{cu} V^c\right)_{v=\beta+\delta\beta} + \int_\beta^{\beta+\delta\beta} dv \left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha} \\
&= + \int_\alpha^{\alpha+\delta\alpha} du \Big[\underbrace{\left(\Gamma^a{}_{cu} V^c\right)_{v=\beta+\delta\beta} - \left(\Gamma^a{}_{cu} V^c\right)_{v=\beta}}_{\approx \delta\beta \frac{\partial}{\partial v}\left(\Gamma^a{}_{cu} V^c\right)_{v=\beta}} \Big] \\
&\quad - \int_\beta^{\beta+\delta\beta} dv \Big[\underbrace{\left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha+\delta\alpha} - \left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha}}_{\approx \delta\alpha \frac{\partial}{\partial u}\left(\Gamma^a{}_{cv} V^c\right)_{u=\alpha}} \Big]
\end{aligned}
$$

$$
\begin{aligned}
&\approx \delta\beta\,\frac{\partial}{\partial v}\big(\Gamma^a{}_{cu}\,V^c\big)_{\substack{u=\alpha\\ v=\beta}}\underbrace{\int_\alpha^{\alpha+\delta\alpha} du}_{\delta\alpha} - \delta\alpha\,\frac{\partial}{\partial u}\big(\Gamma^a{}_{cv}\,V^c\big)_{\substack{u=\alpha\\ v=\beta}}\underbrace{\int_\beta^{\beta+\delta\beta} dv}_{\delta\beta}\\
&= \delta\alpha\,\delta\beta\Big[\frac{\partial}{\partial v}\big(\Gamma^a{}_{cu}\,V^c\big) - \frac{\partial}{\partial u}\big(\Gamma^a{}_{cv}\,V^c\big)\Big]_{\substack{u=\alpha\\ v=\beta}}\\
&= \delta\alpha\,\delta\beta\Big[\big(\partial_v\Gamma^a{}_{cu}\,V^c + \Gamma^a{}_{cu}\underbrace{\partial_v V^c}_{-\Gamma^c{}_{dv}\,V^d}\big) - \big(\partial_u\Gamma^a{}_{cv}\,V^c + \Gamma^a{}_{cv}\underbrace{\partial_u V^c}_{-\Gamma^c{}_{du}\,V^d}\big)\Big]_{\substack{u=\alpha\\ v=\beta}}\\
&= \delta\alpha\,\delta\beta\Big[\partial_v\Gamma^a{}_{cu} - \partial_u\Gamma^a{}_{cv} + \Gamma^a{}_{dv}\,\Gamma^d{}_{cu} - \Gamma^a{}_{du}\,\Gamma^d{}_{cv}\Big]_A\,V^c_A,
\end{aligned}
\tag{6.134}
$$

where $\delta\alpha\,\delta\beta$ is approximately the *area* surrounded by the closed loop. One can find that if the term of the square bracket at point A vanishes, the net change $\Delta V^a_A = 0$, i.e., the vector V'^a_A is parallel to V^a_A. So, such the square bracket term would characterize a geometric quantity and play a crucial role for a space. It is interesting that indices u and v appear antisymmetrically in the square bracket, which implies that the *opposite* transported direction along the closed path leads to different sign of the result.

Generally we can use arbitrary coordinates $\{u^a\}$ to calculate arbitrary closed path c. Follow two approximations

$$
dV^c = -\Gamma^c{}_{ef}\,V^e\,du^f \qquad\Longrightarrow\qquad V^c(u) \approx V^c_A - \Gamma^c{}_{ef}|_A\,V^e_{(A)}\big(u^f - u^f_A\big)
\tag{6.135}
$$

and

$$
\Gamma^a{}_{cb}(u) \approx \Gamma^a{}_{cb}|_A + \Gamma^a{}_{cb,\,d}|_A\big(u^d - u^d_A\big),
\tag{6.136}
$$

the change of V^a_A can be obtained through the integration along the closed path c by

$$
\begin{aligned}
\Delta V^a_A = \oint_c dV^a &= -\oint_c \Gamma^a{}_{cb}(x)\,V^c(u)\,du^b\\
&\approx -\oint_c \Big[\Gamma^a{}_{cb}|_{(A)} + \Gamma^a{}_{cb,\,d}|_A\big(u^d - u^d_A\big)\Big]\Big[V^c_A - \Gamma^c{}_{ef}|_A\,V^e_A\big(u^f - u^f_A\big)\Big]du^b\\
&\approx -\oint_c \Big\{\Big[\Gamma^a{}_{cb}|_A\,V^c_A\Big] + \Big[-\Gamma^a{}_{cb}|_A\,\Gamma^c{}_{ef}|_A\,V^e_A\,\big(u^f - u^f_A\big) + \underbrace{\Gamma^a{}_{cb,\,d}|_A\big(u^d - u^d_A\big)\,V^c_A}_{c\to e\text{ and }d\to f}\Big]\Big\}\,du^b\\
&\approx -\Gamma^a{}_{cb}|_A V^c_A\underbrace{\oint_c du^b}_{0} - \Big[\Gamma^a{}_{eb,\,f}|_A - \Gamma^a{}_{cb}|_A\,\Gamma^c{}_{ef}|_A\Big]V^e_A\underbrace{\oint_c\big(u^f - u^f_A\big)\,du^b}_{\oint_c u^f\,du^b - u^f_A\underbrace{\oint_c du^b}_{0}}\\
&= -\underbrace{\Big[\Gamma^a{}_{eb,\,f}|_A - \Gamma^a{}_{cb}|_A\,\Gamma^c{}_{ef}|_A\Big]}_{[\bullet]_{bf}}V^e_A\oint_c u^f\,du^b.
\end{aligned}
\tag{6.137}
$$

We note that, in the integral, the indices f and b are contracted, so we can rewrite the integral through the trick

$$
\big[\,\bullet\,\big]_{bf}\oint_c u^f\,du^b = \frac{1}{2}\Big\{\big[\,\bullet\,\big]_{bf}\oint_c u^f\,du^b + \big[\,\bullet\,\big]_{fb}\oint_c u^b\,du^f\Big\}.
\tag{6.138}
$$

However, we have $d\big(u^f\, u^b\big) = u^f\, du^b + u^b\, du^f$, the second integral in (6.138) can be replaced:

$$0 = \oint_c d\big(u^f u^b\big) = \oint_c u^f\, dx^b + \oint_c u^b\, du^f \qquad \Longrightarrow \qquad \oint_c u^b\, du^f = -\oint_c u^f\, du^b, \tag{6.139}$$

where the integral can be obtained by

$$\oint_{c=\partial D} u^f\, du^b \underset{\substack{\uparrow \\ \text{By Stokes' theorem}}}{=} \int_D du^f\, du^b := \mathcal{A}^{fb}. \tag{6.140}$$

As a result, the total change of the vector V^a_A is

$$\Delta V^a_A = -\frac{1}{2}\Big[\Gamma^a{}_{eb,\,f}|_A - \Gamma^a{}_{ef,\,b}|_A + \Gamma^a{}_{cf}|_A\,\Gamma^c{}_{eb}|_A - \Gamma^a{}_{cb}|_A\,\Gamma^c{}_{ef}|_A\Big] V^e_A\, \mathcal{A}^{fb}. \tag{6.141}$$

The square bracket is the so-called *Riemann-Christoffel curvature tensor* (or simply the *Riemann curvature tensor* or the *curvature tensor*) at point A

$$\boxed{R^a{}_{efb}|_A := \Big[\Gamma^a{}_{eb,\,f} - \Gamma^a{}_{ef,\,b} + \Gamma^a{}_{cf}\,\Gamma^c{}_{eb} - \Gamma^a{}_{cb}\,\Gamma^c{}_{ef}\Big]_A.} \tag{6.142}$$

It is apparent that $R^a{}_{efb}$ measures the change of the vector V^a at any point within a surface with an area $\mathcal{A}^{fb}$ by

$$\boxed{\Delta V^a = -\frac{1}{2}\big[R^a{}_{efb}\big] V^e\, \mathcal{A}^{fb}.} \tag{6.143}$$

Therefore, if $\Delta V^a \neq 0$ or $R^a{}_{efb} \neq 0$, it indicates that the space is *curved* as Fig. 6.9, otherwise we have a *flat* space as Fig. 6.8 with $\Delta V^a = 0$ or $R^a{}_{efb} = 0$. We note that Fig. 6.8 is a specific case with a *vanishing* affinity ($\Gamma^a{}_{bc} = 0$), i.e.,

$$\Gamma^a{}_{bc} = 0 \text{ everywhere} \qquad \Longrightarrow \qquad R^a{}_{efb} = 0 \text{ everywhere}. \tag{6.144}$$

Flat space

We call a space *flat* if and only if the curvature tensor *vanishes everywhere* ($R^a{}_{efb}(u) = 0$).

A well-constructed example of the effect of the existence of the curvature can be illustrated by a sphere with coordinates $\{u^a\} = \{\theta, \varphi\}$, we focus on a particular tangent vector V^a at its North pole, N. Parallelly transported the vector along the path *counterclockwise* by keeping the directional cosine constant, and finally returned the North pole, N, which is shown in Fig. 6.10. It is apparent that the transported vector V'^a is not parallel to itself before the parallel transport.

However, even if $\Gamma^a{}_{bc} \neq 0$, we can still have a vanishing curvature, i.e.,

$$\Gamma^a{}_{bc} \neq 0 \qquad \Longrightarrow \qquad R^a{}_{efb} = 0. \tag{6.145}$$

A particular example of (6.145) is that the existence of the affinity makes the vector V^a *rotating* around the path but still keeping the angle the same through the parallel transport

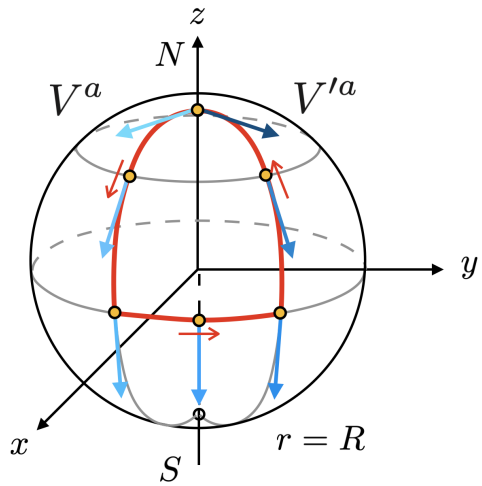


Fig. 6.10: Parallelly transported a vector V^a along a closed loop (red solid line) counterclockwise on a sphere.

along the path, which is shown by Fig. 6.11. Consequencetly, we provide an example to clarify the important concept of the affinity which is always misleading for students that the existence of the affinity leads to a curved space.

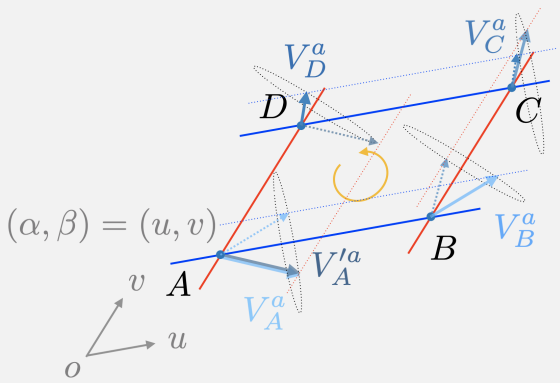


Fig. 6.11: Parallel transport of a vector V^a around a closed loop counterclockwise in a flat space with a non-vanishing affinity.

Note: $\Gamma^a{}_{bc} = 0$ is *not* a condition to determine a flat space. In other words, we can have a *non-vanishing* affinity in the flat space.

Differential relations. If we consider apply the covariant differentiation on a scalar field *twice*

$$\begin{aligned}\boldsymbol{D}\boldsymbol{D}\phi = \boldsymbol{D}d\phi = \boldsymbol{D}(\phi_{,a}\, du^a) &= \underbrace{d\phi_{,a}}_{\phi_{,ac}\, du^c} \otimes du^a + \phi_{,a}(\underbrace{\boldsymbol{D}du^a}_{-\Gamma^a{}_{bc}\, du^c \otimes du^b}) \\ &= \big(\underbrace{\phi_{,ac} - \phi_{,b}\,\Gamma^b{}_{ac}}_{\nabla_c \nabla_a \phi}\big) du^c \otimes du^a. \end{aligned} \tag{6.146a}$$

It can be directly check the result of $\nabla_c\nabla_a\phi$ by expansion

$$\nabla_c\nabla_a\phi = \phi_{;\,ac} = \big(\phi_{;\,a}\big)_{;\,c} = \underbrace{\big(\nabla_a\phi\big)_{,\,c}}_{(\phi_{,\,a})_{,\,c}} - \big(\nabla_b\phi\big)\Gamma^b{}_{ac} = \phi_{,\,ac} - \phi_{,\,b}\,\Gamma^b{}_{ac}. \tag{6.146b}$$

We note that ∇ or the *semicolon* "$;$" in (6.146b) is a short shorthand to denoted two terms (the derivative and connection terms), if we need to use these notation to do the calculation, we need to do the *expansion* on the equation *from outer to inner* since the connection is not a tensor and we do not know how to do the covariant derivative on it. By exchange the indices a and c, we can easily have a similar result:

$$\nabla_a\nabla_c\phi = \phi_{,\,ca} - \phi_{,\,b}\,\Gamma^b{}_{ca}. \tag{6.147}$$

So we obtain an identity

$$\boxed{\big(\nabla_c\nabla_a - \nabla_a\nabla_c\big)\phi = \phi_{,\,b}\big(\Gamma^b{}_{ca} - \Gamma^b{}_{ac}\big) = \phi_{,\,b}\,T^b{}_{ac}.} \tag{6.148}$$

Again, for *Levi–Civita connection*, the right hand side of the identity is vanished:

$$\boxed{\big(\nabla_c\nabla_a - \nabla_a\nabla_c\big)\phi = 0.} \tag{6.149}$$

Now we turn to the case of a vector field $\vec{V} = e_a\,V^a$:

$$\begin{aligned}
\boldsymbol{DD}\vec{V} &= \boldsymbol{DD}\big(e_a\,V^a\big) = \boldsymbol{D}\big[(\underbrace{\boldsymbol{D}e_a}_{e_d\,\Gamma^d{}_{ac}\,du^c})V^a + e_a(\underbrace{dV^a}_{V^a{}_{,\,c}\,du^c})\big] = \boldsymbol{D}\big[(e_d\,\Gamma^d{}_{ac}\,du^c)V^a + e_a(V^a{}_{,\,c}\,du^c)\big] \\
&= \big[(\underbrace{\boldsymbol{D}e_d}_{e_e\,\Gamma^e{}_{db}\,du^b}) \otimes \Gamma^d{}_{ac}\,du^c + e_d\,(\underbrace{d\Gamma^d{}_{ac}}_{\Gamma^d{}_{ac,\,b}\,du^b}) \otimes du^c + e_d\,\Gamma^d{}_{ac}\,(\underbrace{\boldsymbol{D}du^c}_{-\Gamma^c{}_{be}\,du^e\otimes du^b})\big]V^a + (e_b\,\Gamma^b{}_{ac}\,du^c) \otimes \underbrace{dV^a}_{V^a{}_{,\,e}\,du^e} \\
&\quad + \underbrace{\boldsymbol{D}e_a}_{e_e\,\Gamma^e{}_{ab}\,du^b} \otimes (V^a{}_{,\,c}\,du^c) + e_a\big[\underbrace{dV^a{}_{,\,c}}_{V^a{}_{,\,cb}\,du^b} \otimes du^c + V^a{}_{,\,c}\,(\underbrace{\boldsymbol{D}du^c}_{-\Gamma^c{}_{db}\,du^b\otimes du^d})\big] \\
&= e_a\Big\{\big(\Gamma^a{}_{db}\,\Gamma^d{}_{ec} + \Gamma^a{}_{ec,\,b} - \Gamma^a{}_{ed}\,\Gamma^d{}_{cb}\big)V^e \\
&\qquad + 2\,\Gamma^a{}_{eb}\,V^e{}_{,\,c} + V^a{}_{,\,cb} - V^a{}_{,\,d}\,\Gamma^d{}_{cb}\Big\}du^b \otimes du^c.
\end{aligned} \tag{6.150}$$

Now we antisymmetrize the equation above, by procedures

$$\big(du^b \otimes du^c\big)_A \coloneqq du^b \otimes du^c - du^c \otimes du^b, \tag{6.151}$$

and we note

$$\begin{aligned}
K_{bc}\,\big(du^b \otimes du^c\big)_A &= K_{bc}\big(du^b \otimes du^c - du^c \otimes du^b\big) \\
&= K_{bc}\,du^b \otimes du^c - K_{bc}\,du^c \otimes du^b = \big(K_{bc} - K_{cb}\big)du^b \otimes du^c.
\end{aligned} \tag{6.152}$$

Then, we obtain

$$\boxed{\begin{aligned}
(\boldsymbol{DD}\vec{V})_A &= e_a\Big\{\big(R^a{}_{ebc} - \Gamma^a{}_{ed}\,T^d{}_{bc}\big)V^e \\
&\qquad + 2\,(\Gamma^e{}_{ab}\,V^a{}_{,\,c} - \Gamma^e{}_{ac}\,V^a{}_{,\,b}) - V^a{}_{,\,d}\,T^d{}_{bc}\Big\}du^b \otimes du^c \\
&= e_a\Big\{R^a{}_{ebc}\,V^e - V^a{}_{;\,d}\,T^d{}_{bc} + 2\,\big(\Gamma^e{}_{ab}\,V^a{}_{,\,c} - \Gamma^e{}_{ac}\,V^a{}_{,\,b}\big)\Big\}du^b \otimes du^c.
\end{aligned}} \tag{6.153}$$

The covariant derivative twice on $\vec{V}$ is

$$\begin{aligned}
\boldsymbol{D}_b\boldsymbol{D}_c\vec{V} &= \boldsymbol{D}_b\boldsymbol{D}_c\big(e_a\,V^a\big) = \boldsymbol{D}_b\big[(\underbrace{\boldsymbol{D}_c e_a}_{e_d\,\Gamma^d{}_{ac}})V^a + e_a\,V^a{}_{,c}\big] = \boldsymbol{D}_b\big[(e_d\,\Gamma^d{}_{ac})V^a + e_a\,V^a{}_{,c}\big] \\
&= \big[(\underbrace{\boldsymbol{D}_b e_d}_{e_e\,\Gamma^e{}_{db}})\,\Gamma^d{}_{ac} + e_d\,\Gamma^d{}_{ac,\,b}\big]V^a + (e_b\,\Gamma^b{}_{ac})\,V^a{}_{,\,b} + e_e\,\Gamma^e{}_{ab}\,V^a{}_{,\,c} + e_a\,V^a{}_{,\,cb}\big] \\
&= e_a\Big\{\big(\Gamma^a{}_{db}\,\Gamma^d{}_{ec} + \Gamma^a{}_{ec,\,b}\big)V^a + \Gamma^a{}_{ec}\,V^e{}_{,\,b} + \Gamma^a{}_{eb}\,V^e{}_{,\,c} + V^a{}_{,\,cb}\Big\},
\end{aligned} \tag{6.154}$$

and

$$\boldsymbol{D}_c\boldsymbol{D}_b\vec{V} = e_a\Big\{\big(\Gamma^a{}_{dc}\,\Gamma^d{}_{eb} + \Gamma^a{}_{eb,\,c}\big)V^a + \Gamma^a{}_{eb}\,V^e{}_{,\,c} + \Gamma^a{}_{ec}\,V^e{}_{,\,b} + V^a{}_{,\,bc}\Big\}. \tag{6.155}$$

So we have

$$\boxed{\big(\boldsymbol{D}_b\boldsymbol{D}_c - \boldsymbol{D}_c\boldsymbol{D}_b\big)\vec{V} = e_a\,R^a{}_{ebc}\,V^e.} \tag{6.156}$$

It is interesting to apply ∇ twice on the *contravariant* component of $\vec{V}$

$$\begin{aligned}
\nabla_b\nabla_c V^a &= V^a{}_{;\,cb} = \big(V^a{}_{;\,c}\big)_{;\,b} \\
&= \big(V^a{}_{;\,c}\big)_{,\,b} + \Gamma^a{}_{db}\,V^d{}_{;\,c} - V^a{}_{;\,d}\,\Gamma^d{}_{cb} \\
&= \big(V^a{}_{,\,c} + \Gamma^a{}_{ec}\,V^e\big)_{,\,b} + \Gamma^a{}_{db}\,\big(V^d{}_{,\,c} + \Gamma^d{}_{ec}\,V^e\big) - \big(V^a{}_{,\,d} + \Gamma^a{}_{ed}\,V^e\big)\,\Gamma^d{}_{cb} \\
&= V^a{}_{,\,cb} + \Gamma^a{}_{ec,\,b}\,V^e + \Gamma^a{}_{ec}\,V^e{}_{,\,b} + \Gamma^a{}_{db}\,V^d{}_{,\,c} + \Gamma^a{}_{db}\,\Gamma^d{}_{ec}\,V^e \\
&\qquad - V^a{}_{,\,d}\,\Gamma^d{}_{cb} - \Gamma^a{}_{ed}\,V^e\,\Gamma^d{}_{cb} \\
&= V^a{}_{,\,cb} + \Gamma^a{}_{ec}\,V^e{}_{,\,b} + \Gamma^a{}_{db}\,V^d{}_{,\,c} - V^a{}_{,\,d}\,\Gamma^d{}_{cb} \\
&\qquad + \big(\Gamma^a{}_{ec,\,b} + \Gamma^a{}_{db}\,\Gamma^d{}_{ec} - \Gamma^a{}_{ed}\,\Gamma^d{}_{cb}\big)V^e,
\end{aligned} \tag{6.157}$$

and

$$\begin{aligned}
\nabla_c\nabla_b V^a &= V^a{}_{,\,bc} + \Gamma^a{}_{eb}\,V^e{}_{,\,c} + \Gamma^a{}_{dc}\,V^d{}_{,\,b} - V^a{}_{,\,d}\,\Gamma^d{}_{bc} \\
&\qquad + \big(\Gamma^a{}_{eb,\,c} + \Gamma^a{}_{dc}\,\Gamma^d{}_{eb} - \Gamma^a{}_{ed}\,\Gamma^d{}_{bc}\big)V^e,
\end{aligned} \tag{6.158}$$

which results in the well-known *Ricci identity*

$$\boxed{(\nabla_b\nabla_c - \nabla_c\nabla_b)V^a = V^a{}_{;\,cb} - V^a{}_{;\,bc} = R^a{}_{ebc}\,V^e - V^a{}_{;\,d}\,T^d{}_{bc}.} \tag{6.159}$$

Weyl tensor. The trace-free part of the Riemann tensor is defined as the *Weyl tensor*. For n-dimensional space, the Weyl tensor is expressed by

$$C^{ab}{}_{cd} = R^{ab}{}_{cd} - \frac{4}{n-2}\,\delta^{[a}{}_{[c}\,R^{b]}{}_{d]} + \frac{2}{(n-1)(n-2)}\,\delta^{[a}{}_{[c}\,\delta^{b]}{}_{d]}R. \tag{6.160}$$

For $n = 4$ dimensions, it reads as

$$C^{ab}{}_{cd} = R^{ab}{}_{cd} - 2\,\delta^{[a}{}_{[c}\,R^{b]}{}_{d]} + \frac{1}{3}\,\delta^{[a}{}_{[c}\,\delta^{b]}{}_{d]}R$$

$$= R^{ab}{}_{cd} - \frac{1}{2}\Big(\delta^a{}_b\, R^c{}_d - \delta^a{}_b\, R^c{}_d + \delta^c{}_d\, R^a{}_b - \delta^c{}_b\, R^a{}_d\Big)$$
$$+ \frac{1}{6}\Big(\delta^a{}_c\, \delta^b{}_d - \delta^a{}_d\, \delta^b{}_c\Big) R. \tag{6.161}$$

It is apparent that if $R_{\mu\nu} = 0$, $R = 0$ is automatically satisfied, then it results in that the Riemann tensor is fully determined by the Weyl tensor, i.e., $R^{ab}{}_{cd} = C^{ab}{}_{cd}$. Consider a transformation of the line element or the metric *scaling* by a non-negative function $\Omega^2(x)$, called the *conformal transformation*:

$$ds^2 \longrightarrow \widetilde{ds}^2 = \Omega^2(x)\, ds^2 \quad \Longrightarrow \quad g_{ab} \longrightarrow \widetilde{g}_{ab} = \Omega^2(x)\, g_{ab}. \tag{6.162}$$

Hence, it preserves the angle between any two basis vectors. Given a vector $\vec{V}$, such a transformation merely changes the norm of the vector

$$\|\vec{V}\|^2 = g(\vec{V}, \vec{V}) = g_{ab}\, V^a\, V^b \longrightarrow \|\vec{V}\|^2 = \widetilde{g}(\vec{V}, \vec{V}) = \Omega^2(x)\, g_{ab}\, V^a\, V^b. \tag{6.163}$$

It can be shown that, the Weyl tensor is *invariant* under the conformal transformation, i.e.,

$$\boxed{C^{ab}{}_{cd} \longrightarrow \widetilde{C}^{ab}{}_{cd} = C^{ab}{}_{cd}.} \tag{6.164}$$

Because the invariance property under the conformal transformation, the Weyl tensor is also called Weyl's *conformal tensor*.

Gauss-Codazzi equation. Now we have Gauss and Weingarten formulas

$$\begin{cases} \vec{r}_{ac} = \vec{r}_d\, \Gamma^d{}_{ac} + \vec{n}\, b_{ac}, & (6.165a) \\ \vec{n}_b = -\vec{r}_d\, b^d{}_b, & (6.165b) \end{cases}$$

which correspond to the derivative vectors of tangential and normal vectors respectively. By taking the partial derivative of $\vec{r}_{ac}$ with respect to u^b, we have

$$\begin{aligned} \partial_b \vec{r}_{ac} &= \vec{r}_{db}\, \Gamma^d{}_{ac} + \vec{r}_d\, \partial_b \Gamma^d{}_{ac} + \vec{n}_b\, b_{ac} + \vec{n}\, \partial_b b_{ac} \\ &= \big(\vec{r}_e\, \Gamma^e{}_{db} + \vec{n}\, b_{db}\big)\Gamma^d{}_{ac} + \vec{r}_d\, \partial_b \Gamma^d{}_{ac} + (-\vec{r}_d\, b^d{}_b) b_{ac} + \vec{n}\, \partial_b b_{ac} \\ &= \vec{r}_d\big(\partial_b \Gamma^d{}_{ac} + \Gamma^d{}_{eb}\Gamma^e{}_{ac} - b^d{}_b b_{ac}\big) + \vec{n}\big(b_{db}\Gamma^d{}_{ac} + \partial_b b_{ac}\big). \end{aligned} \tag{6.166}$$

By interchanging the indices b and c, we obtain

$$\partial_c \vec{r}_{ab} = \vec{r}_d\big(\partial_c \Gamma^d{}_{ab} + \Gamma^d{}_{ec}\Gamma^e{}_{ab} - b^d{}_c b_{ab}\big) + \vec{n}\big(b_{dc}\Gamma^d{}_{ab} + \partial_c b_{ab}\big). \tag{6.167}$$

We note that

$$\partial_b \vec{r}_{ac} - \partial_c \vec{r}_{ab} = \partial_b \partial_c \vec{r}_a - \partial_c \partial_b \vec{r}_a = 0. \tag{6.168}$$

According to (6.166), (6.167) and (6.168), we obtain a couple of equations called *Gauss-Codazzi equation*

$$0 = \partial_b \vec{r}_{ac} - \partial_c \vec{r}_{ab}$$

$$= \vec{r}_d\big(\underbrace{R^d{}_{abc} - b^d{}_b b_{ac} + b^d{}_c b_{ab}}_{0}\big) + \vec{n}\big(\underbrace{b_{db}\Gamma^d{}_{ac} - b_{dc}\Gamma^d{}_{ab} + \partial_b b_{ac} - \partial_c b_{ab}}_{0}\big) \tag{6.169}$$

due to linearity of $\{\vec{r}_d, \vec{n}\}$, which are usually written as

$$\begin{cases} \textit{Gauss equation}: \boxed{R^d{}_{abc} = b^d{}_b b_{ac} - b^d{}_c b_{ab},} & \text{(6.170a)} \\ \textit{Codazzi equation}: \boxed{\partial_b b_{ac} - \partial_c b_{ab} = b_{dc}\Gamma^d{}_{ab} - b_{db}\Gamma^d{}_{ac}.} & \text{(6.170b)} \end{cases}$$

Here the Riemann(-Christoffel) curvature tensor can also be obtained

$$\boxed{R^d{}_{abc} \coloneqq \partial_b \Gamma^d{}_{ac} - \partial_c \Gamma^d{}_{ab} + \Gamma^d{}_{eb}\Gamma^e{}_{ac} - \Gamma^d{}_{ec}\Gamma^e{}_{ab}.} \tag{6.171}$$

Using the metric tensor, the four lowing-indices curvature tensor can be defined by

$$\boxed{R_{dabc} \coloneqq g_{de} R^e{}_{abc}.} \tag{6.172}$$

We note some symmetries of the Riemann curvature tensor here:

- Antisymmetric in the indices b and c

$$R^d{}_{acb} = -R^d{}_{abc}. \tag{6.173a}$$

- Antisymmetric in d and a (*only* in Riemann space)

$$R_{adbc} = -R_{dabc}. \tag{6.173b}$$

- Symmetric in the pairs of indices (da) and (bc) (*only* in Riemannian space)

$$R_{bcda} = +R_{dabc}. \tag{6.173c}$$

- The cyclic relation (*first Bianchi identity*)

$$R^d{}_{acb} + R^d{}_{cba} + R^d{}_{bac} = 0 \quad \text{or} \quad R^d{}_{[acb]} = 0. \tag{6.173d}$$

We can also contract the first and the third indices in curvature tensor called *Ricci tensor*, which is given by

$$\boxed{R_{ac} = R^b{}_{abc} = g^{db} R_{dabc},} \tag{6.174}$$

and then the *scalar curvature* or *Ricci scalar* is defined by

$$\boxed{\mathbf{R} = R^c{}_c = g^{ac} R_{ac} \overset{\text{or}}{=} g^{ac} g^{db} R_{dabc}.} \tag{6.175}$$

We note the symmetry of the Ricci curvature tensor here:

$$R_{ca} = +R_{ac}. \tag{6.176}$$

Theorema Egregium of Gauss. For *two*-dimensional surface in $\mathbb{R}^3$, the indices $a, b, c, d = 1, 2$, by the properties of Riemann curvature tensor, the following components are vanished:

$$R_{11bc} = R_{22bc} = 0\,, \quad R_{da11} = R_{da22} = 0\,, \tag{6.177}$$

From the Gauss equation (6.170a), the residual one component should uniquely be

$$\begin{aligned} R_{1212} &= b_{11}b_{22} - b_{12}b_{12} \\ &= LN - M^2 \\ &= \det(b_{ab}) \coloneqq \mathbf{b}\,. \end{aligned} \tag{6.178}$$

Therefore, we can rewrite the Gauss curvature (**??**) as

$$\boxed{K = \frac{\mathbf{b}}{\mathbf{g}} = \frac{R_{1212}}{\mathbf{g}}\,,} \tag{6.179}$$

which is a function of g_{ab} *only*, i.e., a 2-dimensional surface in $\mathbb{R}^3$ is *totally* determined by it's intrinsic structure associated with g_{ab}. This is the famous *intrinsic geometry of Gauss* and we call this the *theorema egregium of Gauss*. From the Codazzi equation (6.170b), we only need to consider the case of $a = 1, 2$ and $b = 1$ as well as $c = 2$. This gives

$$\begin{cases} \dfrac{\partial b_{12}}{\partial u^1} - \dfrac{\partial b_{11}}{\partial u^2} = b_{d2}\Gamma^d{}_{11} - b_{d1}\Gamma^d{}_{12}\,, & \text{(6.180a)} \\ \dfrac{\partial b_{22}}{\partial u^1} - \dfrac{\partial b_{21}}{\partial u^2} = b_{d2}\Gamma^d{}_{21} - b_{d1}\Gamma^d{}_{22}\,, & \text{(6.180b)} \end{cases} \quad (d = 1, 2)\,.$$

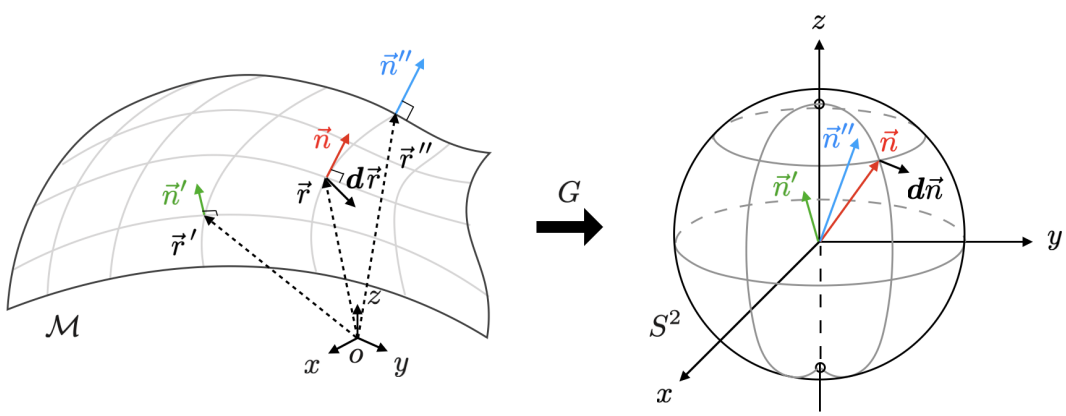


Fig. 6.12: The Gauss map G of a surface.

Third fundamental (quadratic) form. In analogy, we have a Gauss sphere S^2 is shown in Fig. 6.12, which is the image of the Gauss map G from a two-dimensional surface $\mathcal{M}$. A normal vector $\vec{n}$ will be sent to be a radius vector of S^2. Therefore, like the conclusion of (6.13) for a curve in $\mathbb{R}^2$, $\vec{n}$ on S^2 play the same role as $\vec{r}$ on the surface $\mathcal{M}$. As a result, we can calculate the first fundamental for of S^2:

$$\mathrm{I}|_{\mathcal{M}} = d\vec{r} \cdot d\vec{r} \quad \longrightarrow \quad \mathrm{I}|_{S^2} = d\vec{n} \cdot d\vec{n}. \tag{6.181}$$

Now we define the *third fundamental form* of $\mathcal{M}$ to be the first fundamental form of the Gauss sphere S^2

$$\mathrm{III}|_{\mathcal{M}} := \mathrm{I}|_{S^2} = d\vec{n} \cdot d\vec{n}. \tag{6.182}$$

From the first fundamental form, we can calculate the area element on the surface $\mathcal{M}$ and S^2, which are denoted by $d\mathcal{A}|_{\mathcal{M}}$ and $d\mathcal{A}|_{S^2}$ respectively. The results can be obtained by

$$\begin{cases} d\vec{r} \approx \vec{r}_u\, du + \vec{r}_v\, dv & \Longrightarrow \quad d\mathcal{A}|_{\mathcal{M}} = \|\vec{r}_u\, du \wedge \vec{r}_v\, dv\| = \|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv, & (6.183a) \\ d\vec{n} \approx \vec{n}_u\, du + \vec{n}_v\, dv & \Longrightarrow \quad d\mathcal{A}|_{S^2} = \|\vec{n}_u\, du \wedge \vec{n}_v\, dv\| = \|\vec{n}_u \wedge \vec{n}_v\|\, du\, dv. & (6.183b) \end{cases}$$

However, the vectors $\vec{n}_u$ and $\vec{n}_v$ are given by the Weingarten formulas (6.64) with (6.65) and (6.66). We can express $\vec{n}_u \wedge \vec{n}_v$ in terms of $\vec{r}_u$ and $\vec{r}_v$ by

$$\begin{aligned} \vec{n}_u \wedge \vec{n}_v &= AD(\vec{r}_u \wedge \vec{r}_v) - BC(\vec{r}_u \wedge \vec{r}_v) \\ &= \left(\frac{FM - GL}{EG - F^2}\right)\left(\frac{FM - EN}{EG - F^2}\right)(\vec{r}_u \wedge \vec{r}_v) - \left(\frac{FL - EM}{EG - F^2}\right)\left(\frac{FN - GM}{EG - F^2}\right)(\vec{r}_u \wedge \vec{r}_v) \\ &= \frac{LN - M^2}{EG - F^2}(\vec{r}_u \wedge \vec{r}_v) = \frac{\mathbf{b}}{\mathbf{g}}(\vec{r}_u \wedge \vec{r}_v) \underset{\substack{\uparrow \\ \text{By (??)}}}{=} K(\vec{r}_u \wedge \vec{r}_v)\,. \end{aligned} \tag{6.184}$$

Consequently, cf. (6.18), we can measure the absolute value of the Gauss curvature by the ratio

$$\frac{d\mathcal{A}|_{S^2}}{d\mathcal{A}|_{\mathcal{M}}} = \frac{\|\vec{n}_u \wedge \vec{n}_v\|\, du\, dv}{\|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv} = \frac{|K|\|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv}{\|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv} = |K|\,. \tag{6.185}$$

6.6 Integration over a Curved Space

Area element. According to (6.37), we can write down an area element of a surface

$$d\mathcal{A} = \|(\vec{r}_u\, du) \wedge (\vec{r}_v\, dv)\| = \|\vec{r}_u \wedge \vec{r}_v\|\, du\, dv = \|\vec{r}_u\|\|\vec{r}_v\| \sin\angle(\vec{r}_u, \vec{r}_v)\, du\, dv. \tag{6.186}$$

Through the inner product, we can obtain the cosin of the angle $\angle(\vec{r}_u, \vec{r}_v)$

$$\cos\angle(\vec{r}_u, \vec{r}_v) = \frac{\vec{r}_u \cdot \vec{r}_v}{\|\vec{r}_u\|\|\vec{r}_v\|} = \frac{F}{\sqrt{EG}}. \tag{6.187}$$

With the identity $\sin^2\angle(\vec{r}_u, \vec{r}_v) = 1 - \cos^2\angle(\vec{r}_u, \vec{r}_v)$, the *area element* can be obtained

$$d\mathcal{A} = \sqrt{E}\sqrt{G}\sqrt{1 - \frac{F^2}{EG}}\, du\, dv = \sqrt{EG - F^2}\, du\, dv. \tag{6.188}$$

Alternatively one can use the identity for three-dimensional vectors of

$$\boxed{(\vec{A} \wedge \vec{B}) \cdot (\vec{C} \wedge \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{B} \cdot \vec{C})(\vec{A} \cdot \vec{D}),} \tag{6.189}$$

which leads to

$$\|\vec{r}_u \wedge \vec{r}_v\|^2 = (\vec{r}_u \cdot \vec{r}_u)(\vec{r}_v \cdot \vec{r}_v) - (\vec{r}_v \cdot \vec{r}_u)(\vec{r}_u \cdot \vec{r}_v) = EG - F^2. \tag{6.190}$$

As a result, the area of a region D in the surface is

$$\mathcal{A} = \iint_D d\mathcal{A} = \iint_D \sqrt{EG - F^2}\, du\, dv. \tag{6.191}$$

We note that

$$\sqrt{EG - F^2} = \sqrt{g_{11}\, g_{22} - (g_{12})^2} = \sqrt{\mathbf{g}}, \tag{6.192}$$

where we define $\mathbf{g} \coloneqq \det(g_{ab})$. Therefore, the area element can generally be written as

$$\boxed{d\mathcal{A} = \sqrt{\mathbf{g}}\, du\, dv \equiv \sqrt{\mathbf{g}}\, d^2x.} \tag{6.193}$$

Volume element. The total volume can integrate $d\mathcal{V}$ in a region D of a space is given by

$$\mathcal{V} = \iiint_D d\mathcal{V}. \tag{6.194}$$

It can be shown that a *volume element* of a space can be generalized from the area element of (6.193), which should be

$$\boxed{d\mathcal{V} = \sqrt{\mathbf{g}}\, du\, dv\, dw \equiv \sqrt{\mathbf{g}}\, d^3x,} \tag{6.195}$$

if the local coordinates are given by (u, v, w).

Let's consider a n-dimensional Riemannain space $\mathcal{M}$. Let $\{x^i\}$ be the coordinate system. From the formula of the covariant divergence of V^i by analogy to (6.131), we can have a n-dimensional equation of

$$\sqrt{\mathbf{g}}\big(\nabla_i V^i\big) = \partial_i\big(\sqrt{\mathbf{g}}\, V^i\big). \tag{6.196}$$

Doing the integration, we obtain the formula

$$\int_D d\mathcal{V}\, \nabla_i V^i = \int_D d^n x\, \sqrt{\mathbf{g}}\big(\nabla_i V^i\big) = \int_D d^n x\, \partial_i\big(\sqrt{\mathbf{g}}\, V^i\big), \tag{6.197}$$

which becomes a *total divergence term.*

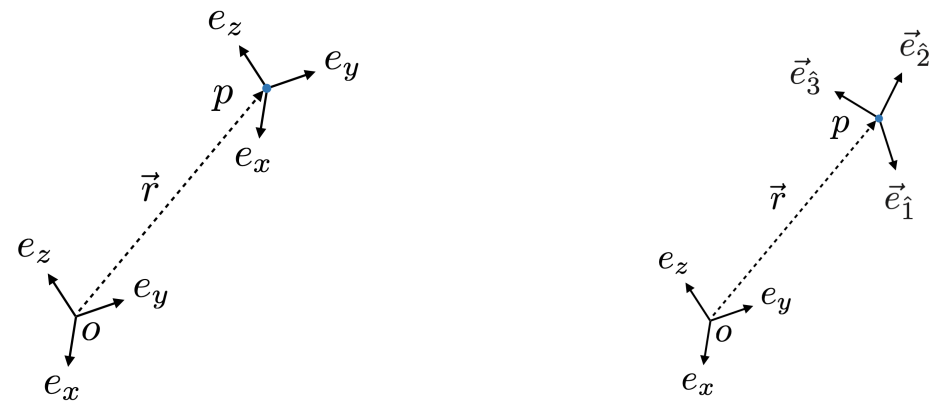
Stokes' theorem. Suppose that a given vector $V = e_i\, V^i$ flows along the direction of e_i, then

$$\boxed{\int_D d^n x\, \sqrt{\mathbf{g}}\, \big(\nabla_i V^i\big) = \oint_{\partial D} d\Sigma_i\, V^i,} \tag{6.198}$$

where

$$d\Sigma_i = \sqrt{\mathbf{h}}\, g_{ij}\, n^j\, d^{n-1}x \tag{6.199}$$

is the surface element of a hypersurface embeded in $\mathcal{M}$ with $\mathbf{h} \coloneqq \det(h_{ab})$ the determinant of the metric tensor of the hypersurface with $n = e_i\, n^i$ the unit normal vector of Σ. For flat space, (6.198) recovers the formula of (3.155).



(a) A Cartesian frame attached on p.

(b) An arbitrary moving frame attached on p.

Fig. 6.13: Local frames at o and p.

6.7 Cartan's Moving Frame Method

A local region of a point o in the space regarded as an Euclidean space $\mathbb{E}^3$ with a reference frame $\{\vec{e}_x, \vec{e}_y, \vec{e}_z\}$. In a region, we can locally attach a Cartesian frame $\{\vec{\delta}_i\} = \{\vec{e}_x, \vec{e}_y, \vec{e}_z\}$ at any point, for example the point p is represented by it position vector $\vec{r}$, which is called the *moving frame* $\{p; \vec{\delta}_i\}$ or $\{\vec{r}; \vec{\delta}_i\}$ in Fig. 6.13(a). Given an infinitesimally nearby point p' represented by its position vector $\vec{r}\,'$, and the attached moving frame would be $\{p'; \vec{e}_i\,'\}$. Now we prefer p as the reference point and then the displacement vector $d\vec{r}$ joining p and p' in terms of $\{\vec{\delta}_i\}$ at p would be

$$\boldsymbol{d}\vec{r} = \vec{\delta}_i\, dx^i = \vec{e}_x\, dx + \vec{e}_y\, dy + \vec{e}_z\, dz. \tag{6.200}$$

Hence, the position vector $\vec{r}\,'$ of p' can be identified by

$$\vec{r} + \boldsymbol{d}\vec{r} = \vec{r} + \vec{e}_x\, dx + \vec{e}_y\, dy + \vec{e}_z\, dz, \quad \text{i.e.,} \quad \boxed{\vec{r}\,' \equiv \vec{r} + \boldsymbol{d}\vec{r}.} \tag{6.201}$$

However, as illustrated by Fig. 6.13(b), we can arbitrarily choose a new basis $\{\vec{e}_{\hat{i}}\} = \{\vec{e}_{\hat{1}}, \vec{e}_{\hat{2}}, \vec{e}_{\hat{3}}\}$

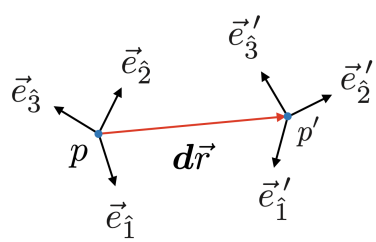


Fig. 6.14: A local displacement vector $\boldsymbol{d}\vec{r}$.

attached at p, which is not necessary orthogonal. So we perform the following transformation for the attached moving frame

$$\boxed{\vec{\delta}_i = \vec{e}_{\hat{i}}\, a^{\hat{i}}{}_i,} \tag{6.202}$$

or more precisely in the matrix form,

$$\mathsf{e} = \bar{\mathsf{e}}\,\mathsf{A} \quad \Longrightarrow \quad \begin{pmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \end{pmatrix} = \begin{pmatrix} \vec{e}_{\hat{1}} & \vec{e}_{\hat{2}} & \vec{e}_{\hat{3}} \end{pmatrix} \begin{pmatrix} a^{\hat{1}}{}_x & a^{\hat{1}}{}_y & a^{\hat{1}}{}_z \\ a^{\hat{2}}{}_x & a^{\hat{2}}{}_y & a^{\hat{2}}{}_z \\ a^{\hat{3}}{}_x & a^{\hat{3}}{}_y & a^{\hat{3}}{}_z \end{pmatrix}. \tag{6.203}$$

Substituting (6.203) into (6.200), the displacement vector becomes

$$\begin{aligned}
\boldsymbol{d}\vec{r} &= \left(e_{\hat{1}}\, a^{\hat{1}}{}_{x} + e_{\hat{2}}\, a^{\hat{2}}{}_{x} + e_{w}\, a^{\hat{3}}{}_{x}\right) dx \\
&\quad + \left(e_{\hat{1}}\, a^{\hat{1}}{}_{y} + e_{\hat{2}}\, a^{\hat{2}}{}_{y} + e_{\hat{3}}\, a^{\hat{3}}{}_{y}\right) dy \\
&\quad + \left(e_{\hat{1}}\, a^{\hat{1}}{}_{z} + e_{\hat{2}}\, a^{\hat{2}}{}_{z} + e_{\hat{3}}\, a^{\hat{3}}{}_{z}\right) dz \\
&= e_{\hat{1}}\left(a^{\hat{1}}{}_{x}\, dx + a^{\hat{1}}{}_{y}\, dy + a^{\hat{1}}{}_{z}\, dz\right) \\
&\quad + e_{\hat{2}}\left(a^{\hat{2}}{}_{x}\, dx + a^{\hat{2}}{}_{y}\, dy + a^{\hat{2}}{}_{z}\, dz\right) \\
&\quad + e_{\hat{3}}\left(a^{\hat{3}}{}_{x}\, dx + a^{\hat{3}}{}_{y}\, dy + a^{\hat{3}}{}_{z}\, dz\right).
\end{aligned} \tag{6.204}$$

Here we have defined a set of new component of $\boldsymbol{d}\vec{r}$ corresponding to the new moving frame $\{\vec{e}_{\hat{1}}, \vec{e}_{\hat{2}}, \vec{e}_{\hat{3}}\}$ in Fig. 6.14:

$$\boxed{\vartheta^{\hat{i}} := a^{\hat{i}}{}_{i}\, dx^{i},} \tag{6.205}$$

and the its matrix form is

$$\bar{\boldsymbol{\vartheta}} = \mathsf{A}\,\boldsymbol{\vartheta} \quad \Longrightarrow \quad \begin{pmatrix} \vartheta^{\hat{1}} \\ \vartheta^{\hat{2}} \\ \vartheta^{\hat{3}} \end{pmatrix} = \begin{pmatrix} a^{\hat{1}}{}_{x} & a^{\hat{1}}{}_{y} & a^{\hat{1}}{}_{z} \\ a^{\hat{2}}{}_{x} & a^{\hat{2}}{}_{y} & a^{\hat{2}}{}_{z} \\ a^{\hat{3}}{}_{x} & a^{\hat{3}}{}_{y} & a^{\hat{3}}{}_{z} \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dx \end{pmatrix}. \tag{6.206}$$

So, cf. (6.200), it generally leads the displacement vector to be the form of

$$\boxed{\boldsymbol{d}\vec{r} = \vec{e}_{\hat{1}}\, \vartheta^{\hat{1}} + \vec{e}_{\hat{2}}\, \vartheta^{\hat{2}} + \vec{e}_{\hat{3}}\, \vartheta^{\hat{3}} \equiv e_{\hat{i}}\, \vartheta^{\hat{i}}} \tag{6.207}$$

and implies that the form of the displacement vector

$$\boxed{\boldsymbol{d}\vec{r} = \mathsf{e}\,\boldsymbol{\vartheta} = (\bar{\mathsf{e}}\,\mathsf{A})(\mathsf{A}^{-1}\bar{\boldsymbol{\vartheta}}) = \bar{\mathsf{e}}\,\bar{\boldsymbol{\vartheta}}} \tag{6.208}$$

is invariant and independent with the choices of frame. Then the line element (square of the length of $\boldsymbol{d}\vec{r}$) between p and q in arbitrary frames can be obtained by

$$ds^2 = g(\boldsymbol{d}\vec{r}, \boldsymbol{d}\vec{r}) = \boldsymbol{d}\vec{r}\cdot\boldsymbol{d}\vec{r} = (\vec{e}_{\hat{i}}\, \vartheta^{\hat{i}})\cdot(\vec{e}_{\hat{j}}\, \vartheta^{\hat{j}}) = (\vec{e}_{\hat{i}}\cdot\vec{e}_{\hat{j}})\, \vartheta^{\hat{i}}\vartheta^{\hat{j}} = g_{\hat{i}\hat{j}}\, \vartheta^{\hat{i}}\vartheta^{\hat{j}}. \tag{6.209}$$

In addition to (6.207), the frame at q can be identified as $\{\vec{e}_{\hat{i}}{}'\} \equiv \{\vec{e}_{\hat{i}} + \boldsymbol{d}\vec{e}_{\hat{i}}\}$. The vectorial variation $\boldsymbol{d}\vec{e}_{\hat{i}}$ in terms of the set of differentials should be

$$\begin{cases} \boldsymbol{d}\vec{e}_{\hat{1}} = \vec{e}_{\hat{1}}\, \omega^{\hat{1}}{}_{\hat{1}} + \vec{e}_{\hat{2}}\, \omega^{\hat{2}}{}_{\hat{1}} + \vec{e}_{\hat{3}}\, \omega^{\hat{3}}{}_{\hat{1}}, \\ \boldsymbol{d}\vec{e}_{\hat{2}} = \vec{e}_{\hat{1}}\, \omega^{\hat{1}}{}_{\hat{2}} + \vec{e}_{\hat{2}}\, \omega^{\hat{2}}{}_{\hat{2}} + \vec{e}_{\hat{3}}\, \omega^{\hat{3}}{}_{\hat{2}}, \\ \boldsymbol{d}\vec{e}_{\hat{3}} = \vec{e}_{\hat{1}}\, \omega^{\hat{1}}{}_{\hat{3}} + \vec{e}_{\hat{2}}\, \omega^{\hat{2}}{}_{\hat{3}} + \vec{e}_{\hat{3}}\, \omega^{\hat{3}}{}_{\hat{3}}. \end{cases} \tag{6.210}$$

Shortly, we can write the set of equations as a tensor equation

$$\boxed{\boldsymbol{d}\vec{e}_{\hat{i}} := \vec{e}_{\hat{j}}\, \omega^{\hat{j}}{}_{\hat{i}},} \quad \text{with} \quad \boxed{\omega^{\hat{j}}{}_{\hat{i}} := \omega^{\hat{j}}{}_{\hat{i}\hat{k}}\, \vartheta^{\hat{k}},} \tag{6.211}$$

which defines the so-called *Euclidean connection* between two frames. If the moving frame $\{p; \vec{e}_{\hat{i}}\}$ is *orthonormal*, then

$$\vec{e}_{\hat{i}} \cdot \vec{e}_{\hat{i}} = \underbrace{\delta_{\hat{i}\hat{i}}}_{\text{(no summation)}} = 1 \quad \text{and} \quad \boldsymbol{d}\vec{e}_{\hat{i}} \cdot \vec{e}_{\hat{i}} + \vec{e}_{\hat{i}} \cdot \boldsymbol{d}\vec{e}_{\hat{i}} = \omega^{\hat{i}}{}_{\hat{i}} + \omega^{\hat{i}}{}_{\hat{i}} = 0 \quad \Longrightarrow \quad \boxed{\omega^{\hat{i}}{}_{\hat{i}} = 0,} \tag{6.212}$$

$$\vec{e}_{\hat{k}} \cdot \vec{e}_{\hat{i}} = \delta_{\hat{k}\hat{i}} = 0 \quad \text{and} \quad \boldsymbol{d}\vec{e}_{\hat{k}} \cdot \vec{e}_{\hat{i}} + \vec{e}_{\hat{k}} \cdot \boldsymbol{d}\vec{e}_{\hat{i}} = \boxed{\omega^{\hat{i}}{}_{\hat{k}} + \omega^{\hat{k}}{}_{\hat{i}} = 0,} \quad (\hat{k} \neq \hat{i}). \tag{6.213}$$

Then (6.210) would reduces to

$$\begin{cases} \boldsymbol{d}\vec{e}_{\hat{1}} = \quad 0 \quad + \vec{e}_{\hat{2}}\,\omega^{\hat{2}}{}_{\hat{1}} + \vec{e}_{\hat{3}}\,\omega^{\hat{3}}{}_{\hat{1}}, \\ \boldsymbol{d}\vec{e}_{\hat{2}} = \vec{e}_{\hat{1}}\,\omega^{\hat{1}}{}_{\hat{2}} + \quad 0 \quad + \vec{e}_{\hat{3}}\,\omega^{\hat{3}}{}_{\hat{2}}, \\ \boldsymbol{d}\vec{e}_{\hat{3}} = \vec{e}_{\hat{1}}\,\omega^{\hat{1}}{}_{\hat{3}} + \vec{e}_{\hat{2}}\,\omega^{\hat{2}}{}_{\hat{3}} + \quad 0. \end{cases} \tag{6.214}$$

Because we have $\mathcal{M} = \mathbb{E}^3$, there is no normal space $\mathcal{M}^\perp$ of $\mathcal{M}$, i.e., $\boldsymbol{d}\vec{e}_{\hat{i}}$ should always be tangent to $\mathcal{M}$ without the normal direction, which implies that $\boldsymbol{d} = \boldsymbol{D}$ is orthogonal projected to itself. So we can directly replace $\boldsymbol{d}$ by $\boldsymbol{D}$ in (6.211)

$$\boldsymbol{D}\vec{e}_{\hat{i}} = \vec{e}_{\hat{j}}\,\omega^{\hat{j}}{}_{\hat{i}\hat{k}}\,\vartheta^{\hat{k}} \equiv (\boldsymbol{D}_{\hat{k}}\vec{e}_{\hat{i}})\vartheta^{\hat{k}}. \tag{6.215}$$

Given any vector $\vec{X} = \vec{e}_{\hat{i}}\,X^{\hat{i}}$, we have

$$\begin{aligned} \boldsymbol{D}\vec{X} &= \boldsymbol{D}(\vec{e}_{\hat{i}}\,X^{\hat{i}}) = (\boldsymbol{D}\vec{e}_{\hat{i}})X^{\hat{i}} + \vec{e}_{\hat{i}}(\boldsymbol{D}X^{\hat{i}}) \\ &= \vec{e}_{\hat{j}}\,\omega^{\hat{j}}{}_{\hat{i}\hat{k}}\,\vartheta^{\hat{k}}X^{\hat{i}} + \vec{e}_{\hat{i}}(X^{\hat{i}}{}_{,\hat{k}})\vartheta^{\hat{k}} = \vec{e}_{\hat{i}}\Big(X^{\hat{i}}{}_{,\hat{k}} + \omega^{\hat{i}}{}_{\hat{j}\hat{k}}X^{\hat{j}}\Big)\vartheta^{\hat{k}}, \end{aligned} \tag{6.216}$$

where we have used the definition of

$$\boldsymbol{D}X^{\hat{i}} = (\boldsymbol{D}_{\hat{k}}X^{\hat{i}})\vartheta^{\hat{k}} = (a_{\hat{k}}{}^{k}\boldsymbol{D}_{k}X^{\hat{i}})\vartheta^{\hat{k}} = (a_{\hat{k}}{}^{k}\,X^{\hat{i}}{}_{,k})\vartheta^{\hat{k}} := (X^{\hat{i}}{}_{,\hat{k}})\vartheta^{\hat{k}}. \tag{6.217}$$

So the covariant derivative of X is

$$\vec{X}' = \frac{\boldsymbol{D}\vec{X}}{ds} = \vec{e}_{\hat{i}}\Big(X^{\hat{i}}{}_{,\hat{k}} + \omega^{\hat{i}}{}_{\hat{j}\hat{k}}\,X^{\hat{j}}\Big)\frac{\vartheta^{\hat{k}}}{ds} = (\boldsymbol{D}_{\hat{k}}\vec{X})\frac{\vartheta^{\hat{k}}}{ds} \quad \text{and} \quad \boxed{\nabla_{\hat{k}}X^{\hat{i}} = (\boldsymbol{D}_{\hat{k}}\vec{X})^{\hat{i}} = X^{\hat{i}}{}_{,\hat{k}} + \omega^{\hat{i}}{}_{\hat{j}\hat{k}}X^{\hat{j}}.} \tag{6.218}$$

If the moving frame is chosen as a coordinate frame, we can make the following replacements:

$$\begin{cases} \{\vec{e}_{\hat{i}}\} & \longrightarrow & \{\partial_i\}, \\ \{\vartheta^{\hat{i}}\} & \longrightarrow & \{dx^i\}, \\ g_{\hat{i}\hat{j}} & \longrightarrow & g_{ij}, \\ \omega^{\hat{i}}{}_{\hat{j}\hat{k}} & \longrightarrow & \Gamma^{i}{}_{jk}. \end{cases} \tag{6.219}$$

The covariant derivative of X is now read by

$$\vec{X}' = \frac{\boldsymbol{D}\vec{X}}{ds} = \vec{e}_{i}\Big(X^{i}{}_{,k} + \Gamma^{i}{}_{jk}\,X^{j}\Big)\frac{dx^k}{ds} = (\boldsymbol{D}_{k}\vec{X})\frac{dx^k}{ds} \quad \text{and} \quad \nabla_{k}X^{i} = (\boldsymbol{D}_{k}\vec{X})^{i} = X^{i}{}_{,k} + \Gamma^{i}{}_{jk}X^{j}. \tag{6.220}$$

Therefore, the *parallel transport* of a vector X gives the following equations in two frames

$$\text{General frame: } \nabla_{\hat{k}} X^{\hat{i}} = (\boldsymbol{D}\vec{X})^{\hat{i}}{}_{\hat{k}} = X^{\hat{i}}{}_{,\hat{k}} + \omega^{\hat{i}}{}_{\hat{j}\hat{k}} X^{\hat{j}} = 0, \tag{6.221a}$$

$$\text{Coordinate frame: } \nabla_k X^i = (\boldsymbol{D}\vec{X})^i{}_k = X^i{}_{,k} + \Gamma^i{}_{jk} X^j = 0. \tag{6.221b}$$

If two frames are associated with the similar relations of (6.202) and (6.205), i.e.,

$$\vec{e}_{\hat{i}} = \vec{e}_i\,(a^{-1})^i{}_{\hat{i}} = \vec{e}_i\,a_{\hat{i}}{}^i \quad \text{and} \quad \vartheta^{\hat{i}} = a^{\hat{i}}{}_i\,dx^i, \tag{6.222}$$

then, from (6.218), we have

$$\begin{aligned}
\vec{X}' &= \vec{e}_i\,a_{\hat{i}}{}^i\Big((a^{\hat{i}}{}_\ell\,X^\ell)_{,\hat{k}} + \omega^{\hat{i}}{}_{\hat{j}\hat{k}}(a^{\hat{j}}{}_j\,X^j)\Big)\frac{a^{\hat{k}}{}_k\,dx^k}{ds} \\
&= \vec{e}_i\,a_{\hat{i}}{}^i\Big(a_{\hat{k}}{}^m(a^{\hat{i}}{}_\ell\,X^\ell)_{,m} + \omega^{\hat{i}}{}_{\hat{j}\hat{k}}\,a^{\hat{j}}{}_j\,X^j\Big)a^{\hat{k}}{}_k\frac{dx^k}{ds}, \\
&= \vec{e}_i\,a_{\hat{i}}{}^i\Big(a_{\hat{k}}{}^m(a^{\hat{i}}{}_{\ell,m}\,X^\ell + a^{\hat{i}}{}_\ell\,X^\ell{}_{,k}) + \omega^{\hat{i}}{}_{\hat{j}\hat{k}}\,a^{\hat{j}}{}_j\,X^j\Big)a^{\hat{k}}{}_k\frac{dx^k}{ds}, \\
&= \vec{e}_i\Big(a_{\hat{i}}{}^i\,a_{\hat{k}}{}^m\,a^{\hat{i}}{}_{\ell,m}\,X^\ell\,a^{\hat{k}}{}_k + \underbrace{a_{\hat{i}}{}^i\,a_{\hat{k}}{}^m\,a^{\hat{i}}{}_\ell\,X^\ell{}_{,m}\,a^{\hat{k}}{}_k}_{X^i{}_{,k}} + a_{\hat{i}}{}^i\,\omega^{\hat{i}}{}_{\hat{j}\hat{k}}\,a^{\hat{j}}{}_j\,X^j\,a^{\hat{k}}{}_k\Big)\frac{dx^k}{ds}, \\
&= \vec{e}_i\Big(X^i{}_{,k} + \underbrace{(\overbrace{a_{\hat{i}}{}^i\,a_{\hat{k}}{}^m\,a^{\hat{i}}{}_{j,m}\,a^{\hat{k}}{}_k}^{a_{\hat{i}}{}^i\,a^{\hat{i}}{}_{j,k}} + a_{\hat{i}}{}^i\,\omega^{\hat{i}}{}_{\hat{j}\hat{k}}\,a^{\hat{j}}{}_j\,a^{\hat{k}}{}_k)}_{\Gamma^i{}_{jk}}X^j\Big)\frac{dx^k}{ds}.
\end{aligned} \tag{6.223}$$

As a result, we obtain the transform of the connection coefficient

$$\boxed{\Gamma^i{}_{jk} = a_{\hat{i}}{}^i\,\omega^{\hat{i}}{}_{\hat{j}\hat{k}}\,a^{\hat{j}}{}_j\,a^{\hat{k}}{}_k + a_{\hat{i}}{}^i\,a^{\hat{i}}{}_{j,k}.} \tag{6.224}$$

It is obvious that if $\vec{e}_{\hat{i}} = \vec{e}_i$ and $\vartheta^{\hat{i}} = dx^i$, i.e., $a^{\hat{i}}{}_i = \delta^{\hat{i}}_i$, then $\omega^{\hat{i}}{}_{\hat{j}\hat{k}} = \Gamma^i{}_{jk}$, two connection coefficients are exactly the same and (6.218) is reduced to (6.220).

Particularly, if we can choose all the moving frame is a *globally* defined by a Cartesian frame, i.e., $\{\vec{e}_{\hat{i}}\} = \{\vec{e}_x, \vec{e}_y, \vec{e}_z\} = \{\vec{\delta}_i\}$, and $\{\vartheta^{\hat{i}}\} = \{dx, dy, dz\} = \{dx^i\}$ in the Euclidean space, then we should have $0 = \boldsymbol{D}\vec{\delta}_i = \vec{\delta}_j\,\Gamma^j{}_{ik}\,dx^k$ for arbitrary $\vec{\delta}_j$ and dx^k, which gives all the connection coefficients are zero ($\Gamma^j{}_{ik} = 0$) on $\mathcal{M}$ and leads $\mathcal{M}$ to be *curvature-free*, i.e., $\mathcal{M}$ is a *flat* space.

Hypersurface. Given a function $z = f(x, y)$, it describes a surface in $\mathbb{R}^3$. It can also be rewritten as the function $g(x, y, z) = z - f(x, y) = 0$. For arbitrary surface, we define a function for a surface by $\Phi(x, y, z)$. Then its unit normal vector can always be given through the gradient of a surface function:

$$\vec{n} := \frac{\vec{\nabla}\Phi(x, y, z)}{\|\vec{\nabla}\Phi(x, y, z)\|} \quad \text{with} \quad \varepsilon := g(\vec{n}, \vec{n}) = 1. \tag{6.225}$$

We can generalized $\mathbb{R}^3$ to a general space $\mathcal{M}$. If we consider a surface Σ in $\mathcal{M}$, the position vector $\vec{r}$ of Σ should be constrained on the tangent space of Σ. Choosing $\{\vec{e}_{\hat{1}}, \vec{e}_{\hat{2}}\}$ to span the tangent space of Σ results in

$$\vec{n} := \vec{e}_{\hat{3}} = \vec{e}_{\hat{1}} \wedge \vec{e}_{\hat{2}}, \tag{6.226}$$

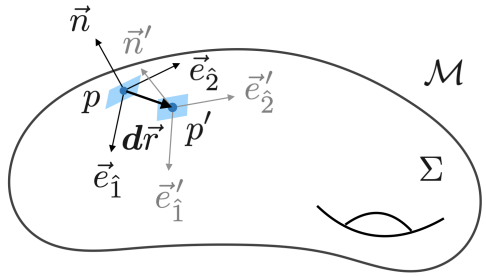


Fig. 6.15: A hypersurface Σ with moving frames $\{p; \vec{e}_{\hat{1}}, \vec{e}_{\hat{2}}, \vec{n}\}$ in $\mathcal{M}$.

and $\{\vec{e}_{\hat{1}}, \vec{e}_{\hat{2}}, \vec{n}\}$ forms a basis in the tangent space at $p \in \mathcal{M}$. Hence, the infinitesimal displacement vector *tangent* to Σ can be expressed by

$$\boldsymbol{d}\vec{r} = \vec{e}_{\hat{1}}\,\vartheta^{\hat{1}} + \vec{e}_{\hat{2}}\,\vartheta^{\hat{2}} + \underbrace{0}_{\vec{n}\,\vartheta^n} := \vec{e}_{\hat{i}}\,\vartheta^{\hat{i}} \quad (\hat{i} = \hat{1}, \hat{2}, n) \qquad \Longrightarrow \qquad \vartheta^n = 0. \tag{6.227}$$

Now (6.214) becomes

$$\begin{cases} \boldsymbol{d}\vec{e}_{\hat{1}} = \quad 0 \quad + \vec{e}_{\hat{2}}\,\omega^{\hat{2}}{}_{\hat{1}} + \vec{n}\,\omega^{n}{}_{\hat{1}}, \\ \boldsymbol{d}\vec{e}_{\hat{2}} = \vec{e}_{\hat{1}}\,\omega^{\hat{1}}{}_{\hat{2}} + \quad 0 \quad + \vec{n}\,\omega^{n}{}_{\hat{2}}, \\ \boldsymbol{d}\vec{n} = \vec{e}_{\hat{1}}\,\omega^{\hat{1}}{}_{n} + \vec{e}_{\hat{2}}\,\omega^{\hat{2}}{}_{n} + \quad 0. \end{cases} \tag{6.228}$$

For m-dimensional hypersurface Σ in $(m+1)$-dimensional $\mathcal{M}$, A frame at $p \in \Sigma$ with an right-handed orientation can be given by $\{\vec{e}_j, \vec{n}\} = \{\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_m, \vec{n}\}$

$$\vec{e}_i = \frac{1}{m!}\,\varepsilon_{ij_1j_2\cdots j_m}\,\vec{e}_{j_1}\,\vec{e}_{j_2}\cdots\vec{e}_{j_m} \quad (j_1, j_2, \ldots, j_m = 1, 2, \ldots, m). \tag{6.229}$$

For $i = m+1$, the normal vector at p is

$$\vec{n} = \varepsilon_{i12\cdots m}\,\vec{e}_1\,\vec{e}_2\cdots\vec{e}_m. \tag{6.230}$$

In general, a vector can be either timelike or spacelike in four-dimensioanl spacetime. Hence, we can generalized to the formula of the sign by

$$\varepsilon := g(\vec{n}, \vec{n}) = \begin{cases} +1 & \text{for spacelike } \vec{n}, \\ -1 & \text{for timelike } \vec{n}. \end{cases} \tag{6.231}$$

6.8 Congruence

The *congruence* means a space-filling family of curves.

6.9 A Few Comments on the Riemannian space

We summary few important properties of the Riemannian space here.

- The Levi–Civita connection coefficients are torsion-free, i.e., symmetry of $\Gamma^k{}_{ij} = \Gamma^k{}_{ji}$, and are introduced through the vectorial variation of an non-fixed frame $e_k\,\Gamma^k{}_{ij} = \boldsymbol{D}_j e_i$.

- The fundamental tensor field is given by the the metric tensor g_{ij} because Levi–Civita connection and Riemann curvature tensor both can be written in terms of g_{ij}.
- The Riemann curvature tensor ${R^k}_{\ell ij}$ is used to check whether or not the space is *flat*, i.e., ${R^k}_{\ell ij} = 0$ for flat space, rather than the determination by either the Euclidean metric tensor δ_{ij} or vanished connection coefficients ${\Gamma^k}_{ij} = 0$.
- Riemannian space is length preserving, i.e., the geometry carries with the metricity condition $\boldsymbol{D}g = 0$ (or $\nabla_k g_{ij} = 0$ in tensor analysis).